

Why Statistics?

- **Without data you are just another person with an opinion** – W Edwards Deming (Data Scientist)
- **It is easy to lie with statistics, it is easier to lie without them** – Frederic Mosteller (Statistician)
- **Statistics is the grammar of science** – Karl Pearson (Statistician)



Simpson's Paradox

- A trend that appears in different groups of data disappears or reverses when the groups are combined
- Example: Between 2000 and 2013, the US median wage went up by about 1% in real terms (i.e. adjusted for inflation)

- Yet:

Group by Education Level	Change in wages
< High school	-7.9%
> High school	-4.7%
Attended University, but no Degree	-7.6%
Degree holders	-1.2%

Simpson's Paradox

- Explanation: Median wage of people with degree went down, but the number of degree holders went up considerably
- Example: Population size = 11

Group by Education Level	Annual Wages	Count
< High school	Rs 5	3
> High school	Rs 10	3
Attended University, but no Degree	Rs 15	3
Degree holders	Rs 20	2

- To calculate median: 5 5 5 10 10 10 15 15 15 20 20 ... Median = 10

Simpson's Paradox

Headline: The national median wage has **gone up** by Rs 4
Reality: For each group, average wage has **declined** by Rs 1

- Next year, the government made a lot of effort to put more students through high school and university, but the average wage of each group fell by Re 1
- Modified data: Population size = 11

Group by Education Level	Annual Wages	Count
< High school	Rs 4	2
> High school	Rs 9	2
Attended University, but no Degree	Rs 14	2
Degree holders	Rs 19	5

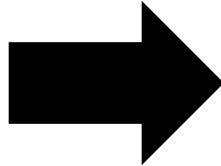
- To calculate median: 4 4 9 9 14 14 19 19 19 19 19 ... Median = 14

Simpson's Paradox – Another Example – Surgery Rate Success at a Hospital

- Group 1 (Young patients):
 - Hospital A: 90 successful surgeries out of 100 (90% success rate)
 - Hospital B: 19 successful surgeries out of 20 (95% success rate)
- Group 2 (Elderly patients):
 - Hospital A: 10 successful surgeries out of 100 (10% success rate)
 - Hospital B: 2 successful surgeries out of 10 (20% success rate)
- Combined Data:
 - Hospital A: 100 successful surgeries out of 200 (50% success rate)
 - Hospital B: 21 successful surgeries out of 30 (70% success rate)

Data Analytics

Current and Historical Data



Transaction ID	Card Number	Amount	Merchant	Time	City
1	1234	4500	AB Hotel	13.50	Pune
2	5678	817	PQ Petrol Pump	08.23	Mumbai
3	9012	9092	RS Shop	15.21	Chennai
4	1890	345	TM Store	10.03	Hyderabad
..					
..					
99999999	5632	82000	Amazon.in	19.11	Pune
100000000	8971	100	Subzi.com	09.36	Bangalore

Using Data



Valid transaction?

Invalid transaction?

Amount range: 1-5000

Amount range: 5001-10000

Amount range: 10000-25000

Amount range > 25001

Time: Morning

Time: Afternoon

Time: Evening

Time: Night

City: Pune

City: Mumbai

City: Delhi

City: Other

Reason: Shopping

Reason: Hotel

Reason: Bill Pymt

Reason: Online



Clusters

Operational Data (Data Comes In)

- An account holder transfers money
- A customer places an order
- A loan is granted



Analytical Data (Data Goes Out)

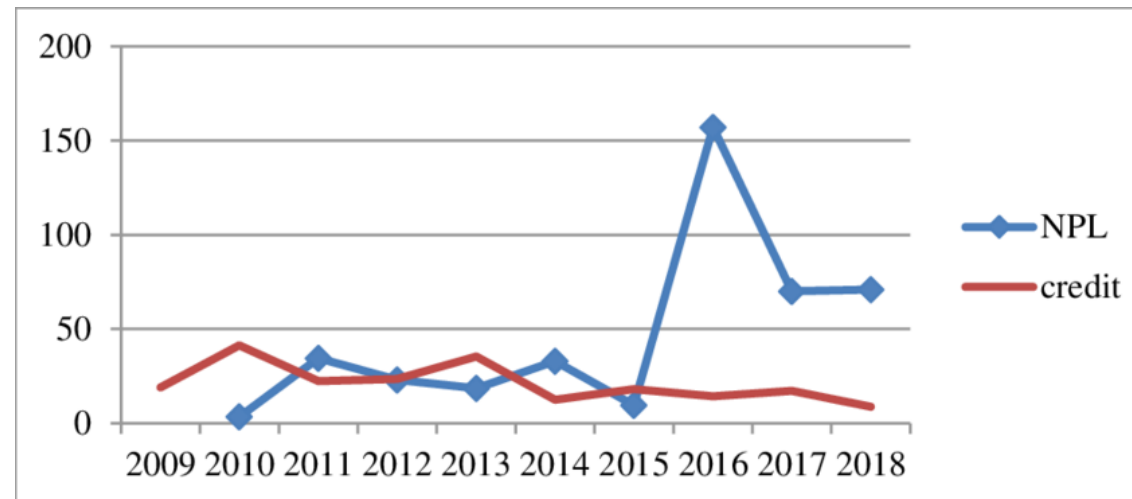
Txn Date	Value Date	Description	Ref No/Cheque No	Debit	Credit	Balance
02 Jan 2019	02 Jan 2019	XXXXXXX	XXXXXXX	XXXXX	XXXXX	XXXXXXX
10 Jan 2019	10 Jan 2019	XXXXXXX	XXXXXXX	XXXXX	XXXXX	XXXXXXX
15 Jan 2019	15 Jan 2019	XXXXXXX	XXXXXXX	XXXXX	XXXXX	XXXXXXX
20 Jan 2019	20 Jan 2019	XXXXXXX	XXXXXXX	XXXXX	XXXXX	XXXXXXX
25 Jan 2019	25 Jan 2019	XXXXXXX	XXXXXXX	XXXXX	XXXXX	XXXXXXX

- Last month money transfer history

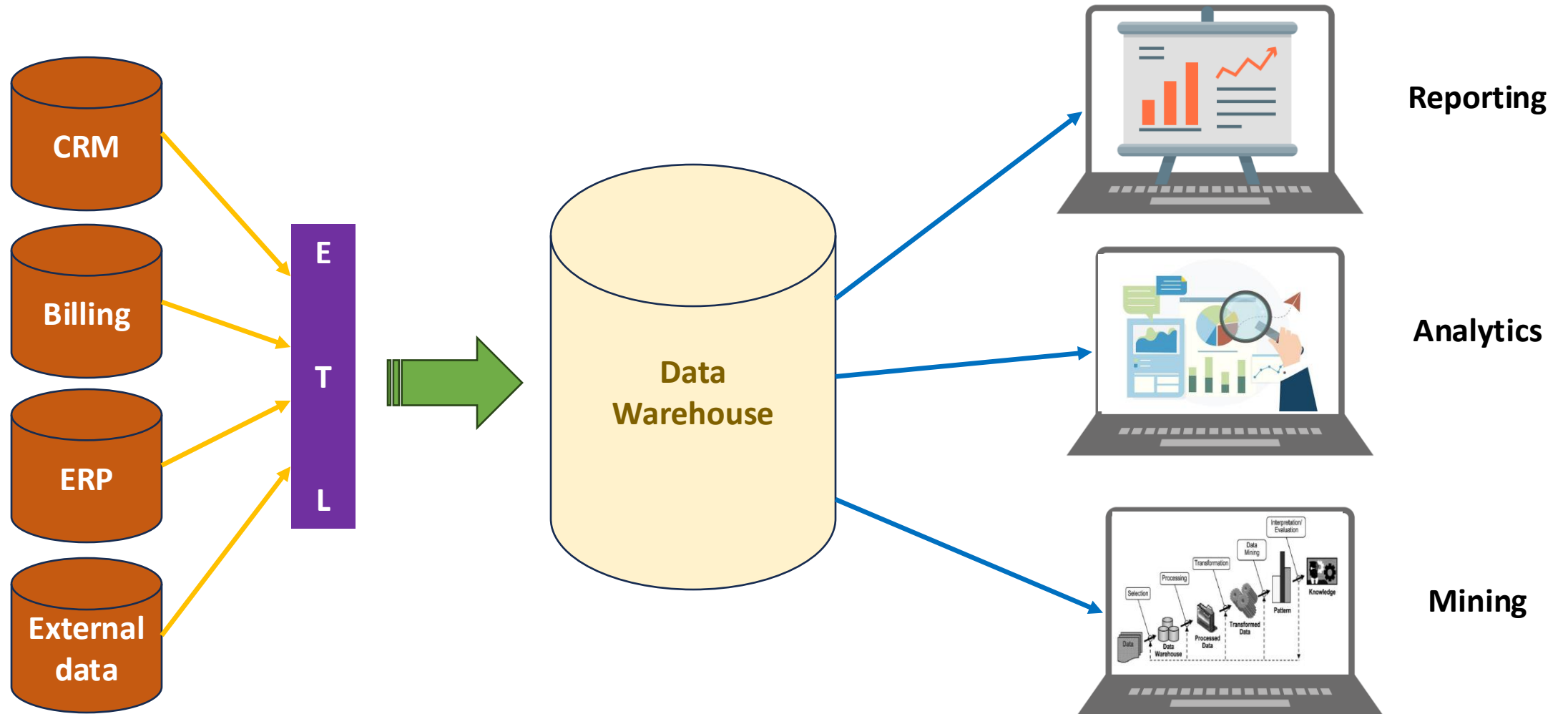
Top 5 Sales Person by Order Conversion %			
SalesPerson	Enquiries	Orders	Order Conv %
Dion	1	1	100%
Judy	1	1	100%
Laura	1	1	100%
Kyle	57	24	42%
Shaz	10	4	40%
Total	70	31	44%

- Top 5 order conversions

- Loans report



Data Warehousing to Data Analytics



Areas of Data-related Work

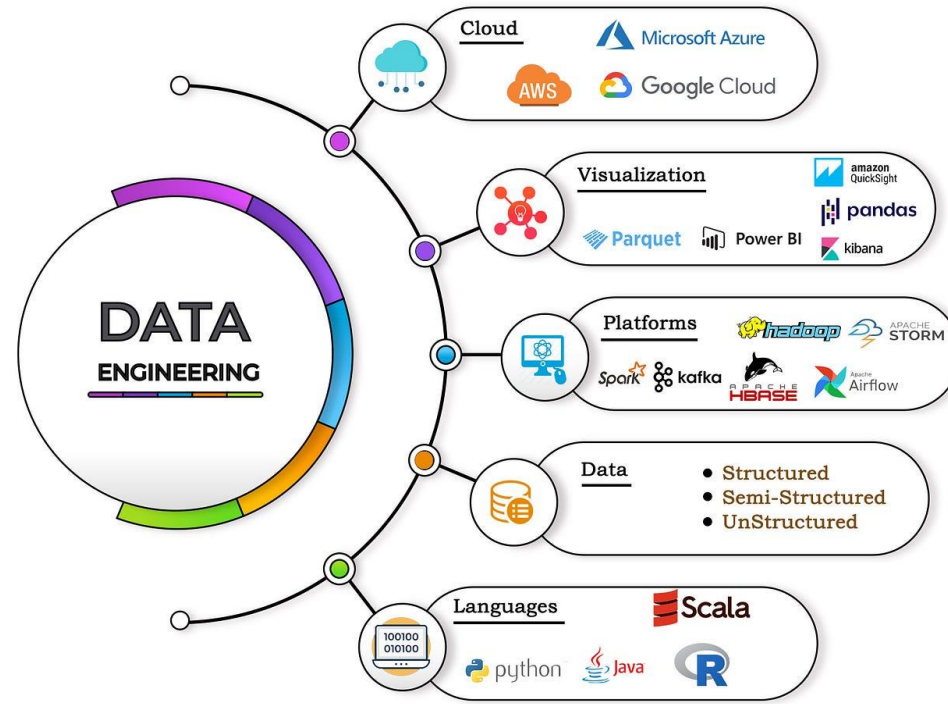
Data – What is There for Me?

Data
Engineering
(Infrastructure)

Data
Analytics
(Decisions)

Data
Science
**(Modelling,
Product building)**

Data Engineering



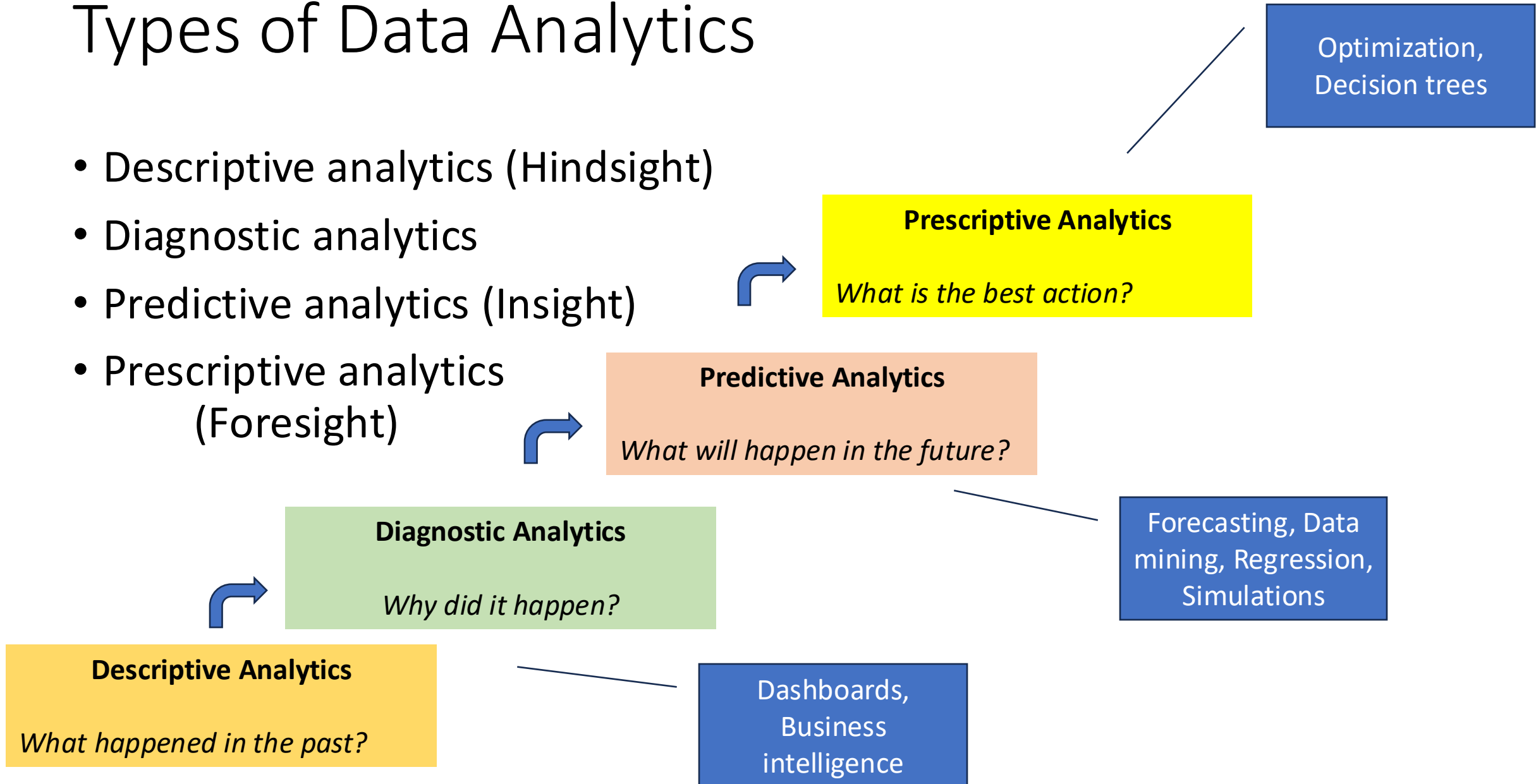
- **Focus:** Creating and maintaining infrastructure for Data Analysts and Data Scientists
- **Skills:** Programming (Python/R), Apache Spark, Kafka, Databases (SQL/MongoDB), Linux Shell Scripting, Cloud (AWS), DevOps (Docker, Kubernetes, Jenkins, Git), ETL

Data Analytics

- What happened?
 - What were the sales figures?
 - How many users registered?
 - How many subscriptions got cancelled?
- Why did it happen?
 - Which products were sold the most?
 - Did we have more mobile users?
 - Did cancellations happen after watching a particular movie?
- What is likely to happen?
 - What will the sale be in this quarter?
 - How many total users will we have?
 - Will subscription cancellation rate drop to 10%?
- What action should we take?
 - How do we grow sales?
 - How do we get to 1M users?
 - How do we take the subscription cancellation rate to 5%?

Types of Data Analytics

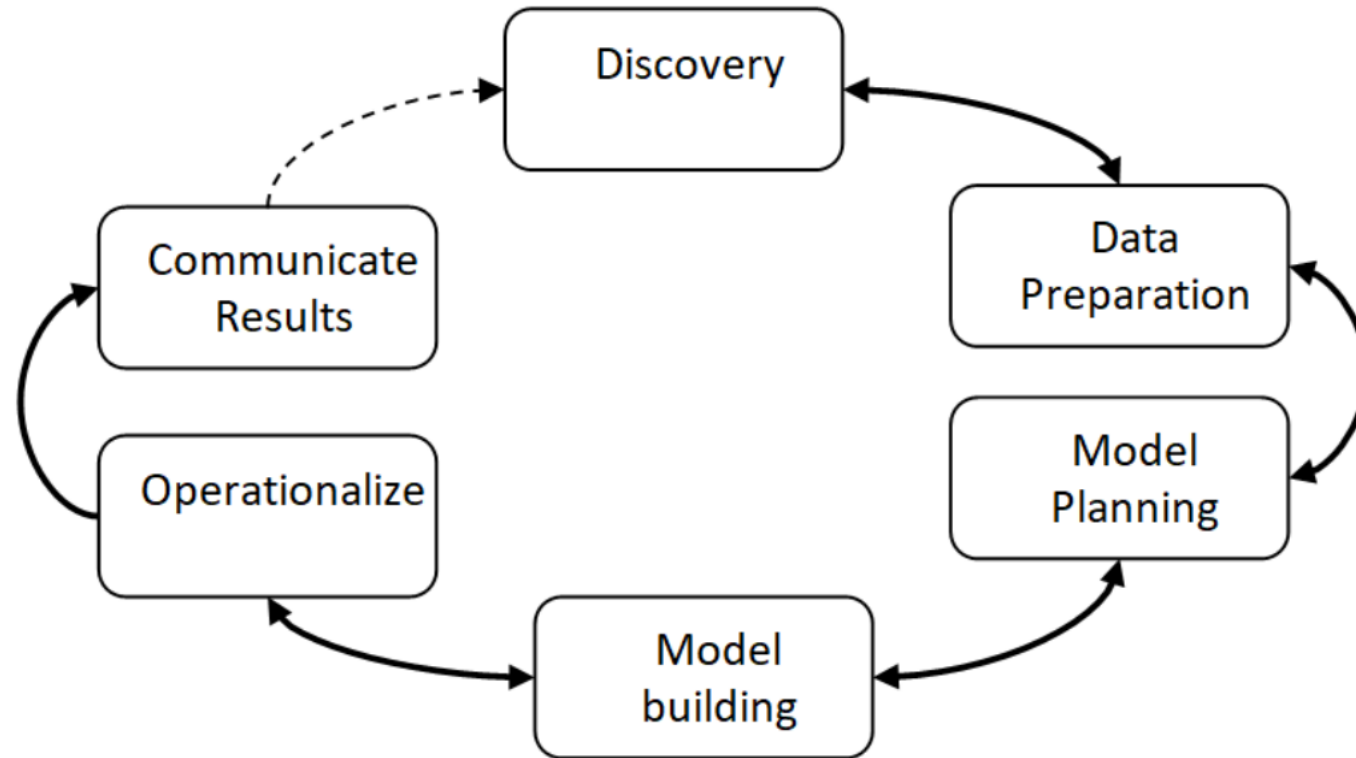
- Descriptive analytics (Hindsight)
- Diagnostic analytics
- Predictive analytics (Insight)
- Prescriptive analytics (Foresight)



Data Analytics Types

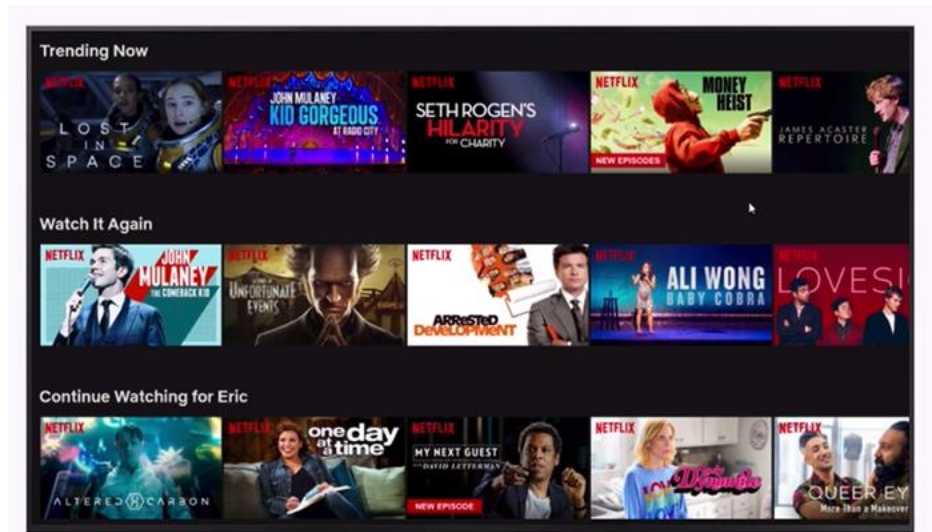
Type	Description	Examples
Descriptive	The use of data to understand past and current business performance and make informed decisions	Which product had the maximum ups and downs in sale last year? Which students scored very high or very low marks?
Diagnostic	Why something happened?	How much salary should we offer to this prospective employee? (Regression analysis)
Predictive	Predict the future by examining historical data, detecting patterns or relationships in these data, and then extrapolating these relationships forward in time.	Predict whether a customer will churn at HDFC Bank, Predict next year sale for Walmart
Prescriptive	Use optimization to identify the best alternatives to minimize or maximize some objective	Maruti Suzuki should manufacture x SUVs, y hatchbacks, z commercial vehicles every three days to maximize its profit

Data Analytics Life Cycle



Data Science

- **Focus-1:** Advanced statistical modelling (e.g. Machine learning, Time series analysis, Decision tree-based techniques, Bayesian networks, etc) – Focus is always on the future
- **Focus-2:** Build data products



Python Data Analytics/Science:
NumPy, Pandas, Matplotlib,
Seaborn, ScikitLearn (SkLearn)

NumPy

- **NumPy**: Base for everything in Python Data Analytics
- Python library for creating N-dimensional arrays
- Similar to Python lists, but much more efficient
- Ability to quickly broadcast functions (No need to do element-by-element operations on arrays)
 - Example:
 - `arr = np.array([1, 2, 3, 4])`
 - `scalar = 10`
 - `result = arr + scalar`
- Built-in linear algebra, statistical distributions, random number capabilities

NumPy Arrays

- Main ways of creating a NumPy array:
 - Transform Python list
 - Use built-in functions
 - Generate random data
- Indexing and Selection: Single element, Slicing, Broadcasting, 2D indexing and selection, Conditional selection
- Operations on arrays: Arithmetic, Universal functions, Summary statistics, 2D arrays
- Code: C:\code\Data Analytics\numpy

Series (C:\code\Data Analytics\pandas\1_pandas_series.py)

- **Series:** A data structure that holds an array of information along with a named index
- The named index distinguishes it from a NumPy array

**NumPy array has
numeric index**

Index	Data
0	1776
1	1867
2	1821

Finding data using this index is not easy

**Pandas series has a
labelled index**

Labelled Index	Data
USA	1776
Canada	1867
England	1821

Finding data using this index is very easy

**Note: Data is internally still
numerically organized!**

Numeric Index	Labelled Index	Data
0	USA	1776
1	Canada	1867
2	England	1821

We can still use the numeric index, if we want

DataFrame

- **DataFrame:** Table of columns and rows that can be easily restructured/filtered

Series

Index	Year
USA	1776
Canada	1867
England	1821

Multiple Series with the Same Index

Index	Year
USA	1776
Canada	1867
England	1821

Index	Pop
USA	328
Canada	38
England	126

Index	GDP
USA	20.5
Canada	1.7
England	3.9

Dataframe

Index	Year	Pop	GDP
USA	1776	328	20.5
Canada	1867	38	1.7
England	1821	126	3.9

- So, Dataframe = Several series that share the same index, like a spreadsheet

DataFrame

- Basic operations
 - Create a dataframe
 - Select a column/multiple columns
 - Select a row/multiple rows
 - Insert a new column/row
- Advanced Operations
 - Indexing
 - Filtering
 - Missing data
 - Grouping
 - Joining

Matplotlib

- **Data visualization** is very important to quickly understand trends and relationships in the data
- **Matplotlib**: One of the most popular plotting libraries in Python
- *Grandfather* of plotting and visualization libraries in Python
- Seaborn/Pandas built-in visualization are built on top of Matplotlib
- Heavily inspired by the plotting functions in the MatLab programming language
- Almost any type of plot can be created and customized
- Can be confusing at first: Two approaches: (1) Functional (2) OOP
- Matplotlib basic plots: Functional, Figures and subplots: OOP
- Main goals
 - Plot a functional relationship, e.g. $y = 2x$
 - Plot a relationship between raw data points: $x = [1, 2, 3, 4]$ and $y = [2, 4, 6, 8]$

Seaborn

- **Seaborn:** Statistical plotting library specifically designed to interact with Pandas DataFrames to create common statistical plots
- Built on top of Matplotlib, but uses a simpler *one-line* syntax
- Easy to use, but less customization possible as compared to Matplotlib
- Types of plots
 - Scatter plots: How are different features related to one another?
 - Distribution plots: How is data distributed?
 - Categorical plots
 - Comparison plots
 - Seaborn grids
 - Matrix plots

Data Visualization Summary (C:\code\Data Analytics\visualization.py)

Plot	Usage	Example	Code
Line plot	Trends over time periods/data points	Stock prices over a month	<code>plt.plot([1, 2, 3], [4, 5, 6]); plt.show()</code>
Scatter plot	Relationship between two numeric variables	House price versus Area of the house	<code>plt.scatter([1, 2, 3], [4, 5, 6]); plt.show()</code>
Bar plot	Compare categories/groups with respect to numeric values	Sales across product categories	<code>plt.bar(['A', 'B', 'C'], [4, 7, 1]); plt.show()</code>
Histogram	Distribution of a single numeric variable	Distribution of ages in a country	<code>plt.hist([1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5], bins=5); plt.show()</code>
Box plot	Distribution of data with reference to minimum, Q1, Q2, Q3, maximum	Income distribution across professions	<code>plt.boxplot([7, 2, 5, 13, 9, 6]); plt.show()</code>
Heatmap	Matrix to display values using colour intensity	Correlation matrix between height and weight	<code>sns.heatmap(np.random.rand(5, 5), annot=True); plt.show()</code>
Pie chart	Proportions/percentages among categories	Market share of mobile phone brands	<code>plt.pie([40, 30, 20, 10], labels=['A', 'B', 'C', 'D'], autopct='%1.1f%%'); plt.show()</code>

Statistics: Basic Concepts

Individuals and Variables

- **Individual:** Item in a data set
- **Variable:** Property of an individual
- **Independent variable (Feature):** Used to predict something
- **Dependent variable:** The predicted value

Person	Hours of Study	Grade
Alice	100	B
Bob	130	A
Charlie	60	E
Dave	90	C

Person: Individual

Variable: Hours of Study, Grade

Independent variable/Feature: Hours of Study

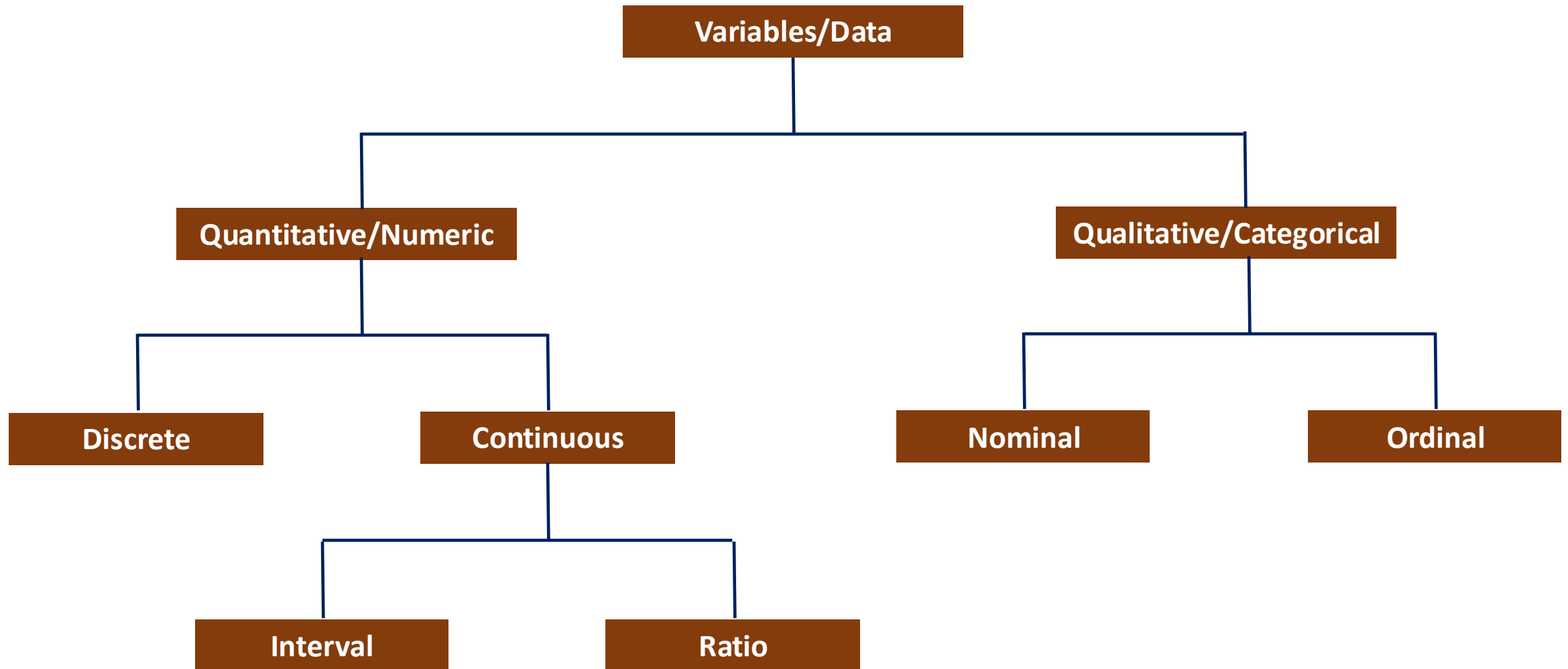
Dependent variable: Grade

Variable Types: Qualitative and Quantitative

- **Qualitative variables: Categorical** (e.g. Grade), **Non-numeric**
- **Quantitative variables: Numeric**, Can be measured (e.g. Hours of Study)

Person	Hours of Study	Grade
Alice	100	B
Bob	130	A
Charlie	60	E
Dave	90	C

Variable Classification

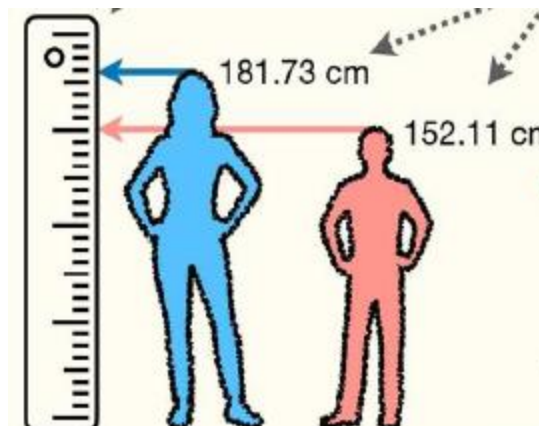
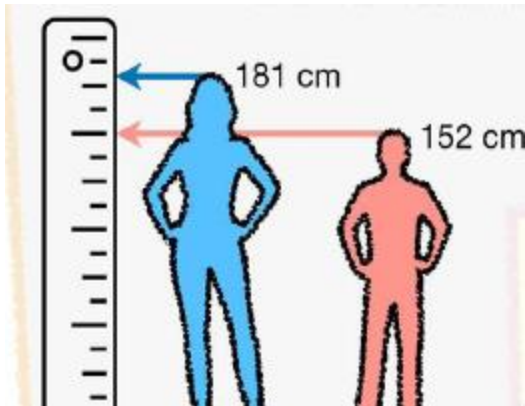


Quantitative Variable Sub-types

- **Discrete variables:** Countable and take specific values ... Example:
Rank: 1, 2, 3 but not 1.8



- **Continuous variables:** Measurable, but not a specific value ...
Example: Height of a person: 5 feet 5 inches 5 cm 2 mm



Levels/Scales of Measurement of Data

Level	Details	Examples
Nominal	Categorical data with no inherent order or ranking	Gender, Colour, Cuisine
Ordinal	Categorical data with a meaningful order or ranking, but intervals between ranks are inconsistent or unknown	Education level (High school, Bachelor's, Master's), Customer satisfaction (Satisfied, Neutral, Unsatisfied)
Interval	Numeric data with meaningful and consistent intervals between them without a true zero point ... Ratio cannot be measured	IQ, Calendar data, Time of the day, Temperatures in Celsius or Fahrenheit
Ratio	Numeric data where both intervals and ratios are meaningful and there is a true zero point	Height, Weight, Income, Distance travelled, Student marks

True zero point: Does the value of 0 actually indicate the start?

Ratio data: Height of 0 inches means 0 inches, or income of 0 means \$0, etc ... Nothing can be lower than this ... (0 is NOT arbitrary)

Interval data: Temperature of 0 degrees Celsius does not mean the lowest temperature, as we can have -1 degrees Celsius, etc ... (0 is arbitrary)

Ratios

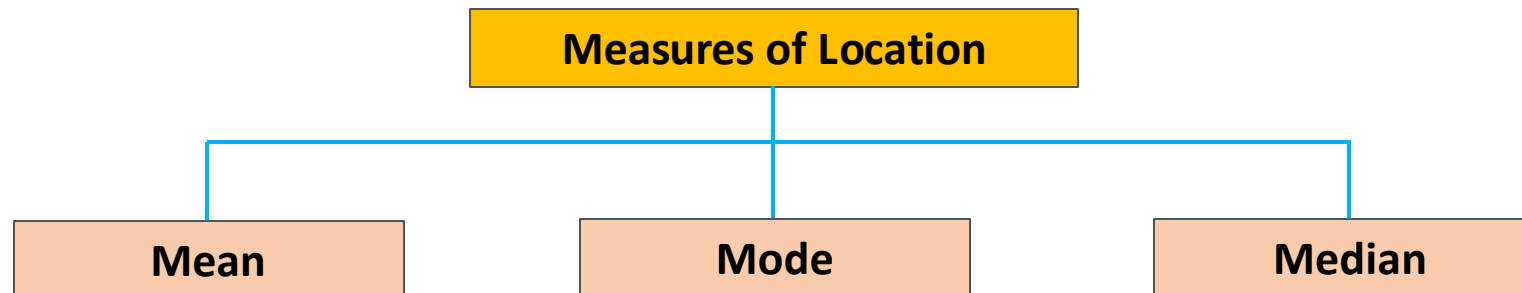
Is a person with IQ 150 *three times* more intelligent than a person with IQ 50? **No!**

Has a student who has got 90 marks done *three times better* than a student who has got 30 marks? **Yes!**

Descriptive Statistics

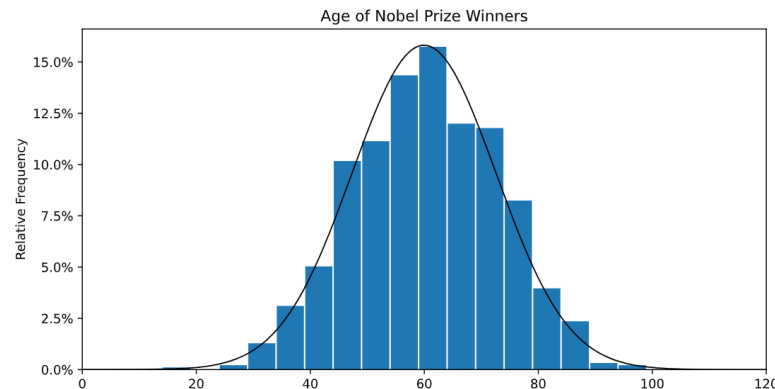
Measures of Location

- **Measures of Location:** A single value that represents the “centering” of a set of data, e.g. average
- Also called **Measures of Central Tendency**
- Example: Marks obtained by 10 students, arranged in an ascending order ... 45,56,61,65,68,71,73,79,82,88,91
- Possible measure of location: 45,56,61,65,68, **71**, 73,79,82,88,91

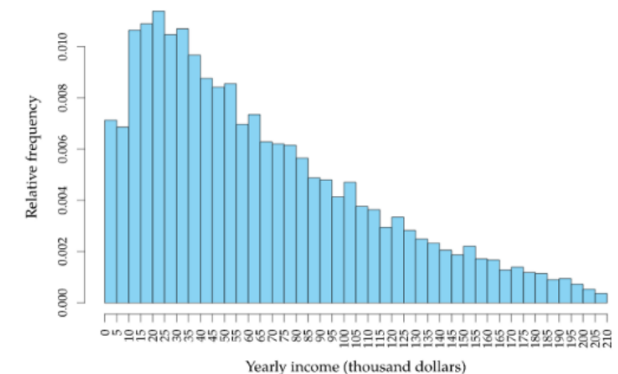


Basic Usage

- **Mean:** Better if the data is **normally distributed** and there are no **outliers** ... Used for interval and ratio data
- **Median:** Better when the data is **skewed** (has extreme values) ... Used for ordinal, interval, and ratio data
- **Mode:** Useful for identifying the most common value or values in a dataset ... Used in all the four scales ... Best for categorical data



Normally distributed data



Skewed data

Mean

- **Mean:** Average
- Sample Marks: 45,56,61,65,68,71,73,79,82,88,91
- Mean = $\frac{45+56+61+65+68+71+73+79+82+88+91}{11}$
- Mean = $\frac{789}{11}$
- Mean ≈ 71.727
- In statistical terms: $\mu = \frac{\sum_{i=1}^n x_i}{n}$
- Extreme values (**outliers**) can impact mean
 - Test scores: 70, 80, 90, 85, and 75 ... Mean: 80
 - Test scores: 70, 80, 90, 85, 75, and 2 ... Mean: 66

Median

- **Median:** The *middle* value when values are in ascending/descending order
- Divides the dataset into two equal halves
 - Odd number of values in the dataset: Median is the middle value
 - Even number of values in the dataset: Median is the average of the two middle values
- Scores: 62, 78, 84, **89**, 91, 95, 97 ---> Median = 89
- Scores: 62, 78, **84**, **89**, 91, 95 ---> Median = $\frac{84+89}{2} = 86.5$
- Outliers do not impact median as much as they impact mean
 - Mean of 56, 78, 45, 49, 55, 62 = 57.5
 - Mean of 56, **180**, 45, 49, 55, 62 = 74.5
 - Corresponding medians = 55.5 and again = 55.5

Mode

- **Mode:** The value that occurs most frequently in a dataset
- Data: 62, 78, 84, 89, 91, 95, 97, 89, 91, 89
- Frequency: 62: 1, 78: 1, 84: 1, **89: 3**, 91: 2, 95: 1, 97: 1
- Mode = 89
- What if there are multiple values with the same highest frequency?:
Multimodal data
 - If we have two modes: **bi-modal**
 - If we have three modes: **tri-modal**
- Not used much in practice

Feature Engineering

- **Feature engineering:** Transform raw data into meaningful **features**
- Why? Improve the performance of the machine learning models
- How?
 - Create new columns (From Date of purchase, create weekday/weekend)
 - Scale features (Bring features on the same scale, e.g. age and income)
 - Encode categorical features (Gender: Convert F = 0, M = 1), since ML models work with numeric data
 - Handle missing data (Drop, Indicate using a *Missing* flag, or **Impute** with mean/mode/median)
 - Feature selection (Keep only the most relevant features)
 - Feature interaction (From *unit price* and *quantity*, create *bill amount*)

Feature Engineering Example

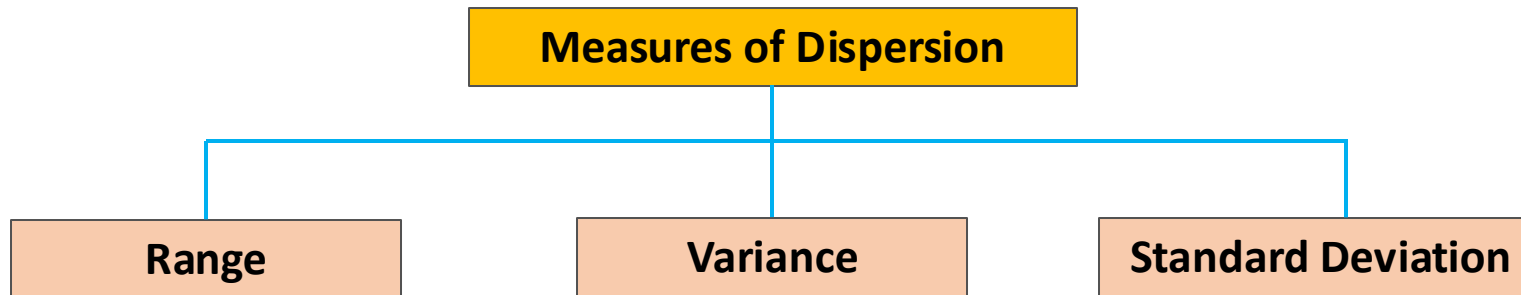
- Consider some sample data

Age	Weight	Salary (K)	Gender	Degree
24	70	40	M	BE
25	65	50	F	BTech
27	85	65	F	-
-	72	45	-	BE
25	-	55	F	BTech

- How to handle missing data?
 - Plot all age values on a graph and see if you see symmetry
 - If yes, impute missing values with mean, else with median: Repeat for others

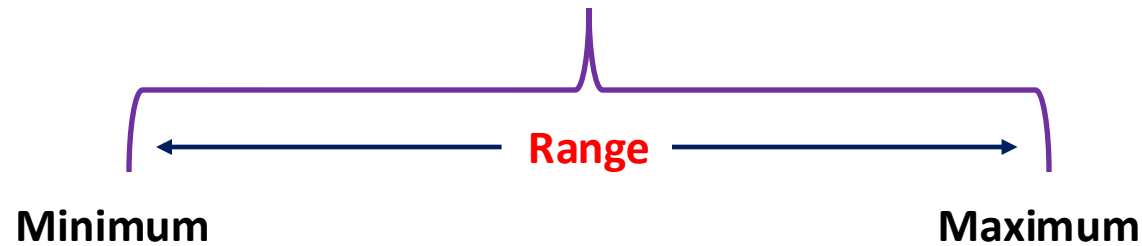
Measures of Dispersion

- **Spread:** How and by how much, our data set is *spread out* around its center?
- Called **Measures of Dispersion** or **Scatter**



Range

- **Range:** Difference between the maximum value and the minimum value in the data set
- Affected by outliers



- Example: 8, 11, 5, 9, 7, 6, 2500
- $\text{Range} = \text{Max} - \text{Min} = 2500 - 5 = 2495$, which is quite meaningless
- Solution: **Inter Quartile Range (IQR)**
- But first, we need to understand **percentile** and **quartile**

Percentile

- **Percentile** (Relative): $\neq$ Percentage (Absolute)
- Percentile: A value below which certain percentage of observations lie
- Slices percentage data into two parts: Below a certain cut off, Above the same cut off
- k^{th} percentile = $k\%$ data is below it, and rest is above it
- Examples:
 - If you are in the 90^{th} percentile in an examination, 90% students are below you and 10% students are above you
 - If a patient's blood pressure is in the 60^{th} percentile, 60% patients have a blood pressure less than this patient, and 40% patients have higher blood pressure than this patient
- Median = 50^{th} percentile

Percentile Example

In some references, we might see *Number of observations*, rather than *Number of observations + 1* ...
Generally does not make a big difference

- Sorted Marks of 20 students:
65,72,78,80,81,83,85,87,88,90,91,92,93,94,95,96,97,98,99,100
- 10th percentile position = $\frac{\text{Desired percentile}}{100} \times \text{Number of observations} + 1 = \frac{10}{100} \times 21 = 2.10$
- Marks at 10th percentile = Marks at the second position $\sim \frac{(72 + 78)}{2} = 75$
- Interpretation: 10% students lie below 72 marks, 90% are above 72 marks
- 90th percentile position = $\frac{\text{Desired percentile}}{100} \times \text{Number of observations} + 1 = \frac{90}{100} \times 21 = 18.9 \sim \frac{(98 + 99)}{2} = 98.5$
- Marks at 90th percentile = Marks at the 18th position = 98
- Interpretation: 90% students lie below 98 marks, 10% are above 98 marks

Percentile Example

- Sorted Marks of 20 students:
65,72,78,80,81,83,85,87,88,90,91,92,93,94,95,96,97,98,99,100
- Another type of question: At what percentile are 85 marks in the above list?
- Percentile for 85 marks = $\frac{\text{Number of observations below the given value}}{\text{Number of observations}} \times 100 = \frac{6}{20} \times 100 = 30$
- 85 marks are at the 30th percentile
- Interpretation: 30% students lie below 85 marks, 70% are above 85 marks
- Note: Here we use *Number of observations*, and not *Number of observations + 1*

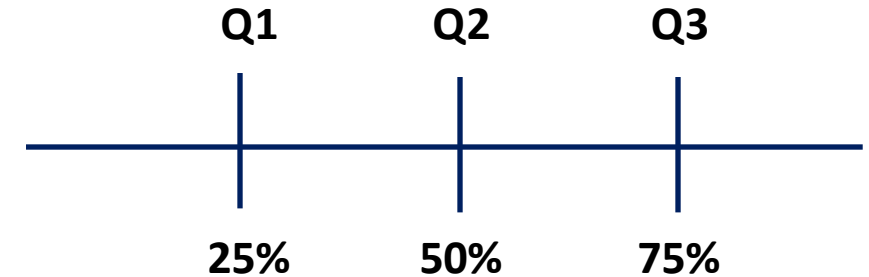
US Household Net Worth and Percentile

(Source: <https://finance.yahoo.com/news/wealthy-net-worth-considered-poor-190014440.html>)

Category	Percentile	Net Worth
Poor	20 th	\$10,000
Middle class	50 th	\$281,000
Wealthy	90 th	\$1.9 million

Quartile

- Percentile: Divides data into 100 equal parts
- **Quartile**: Divides data into 4 equal parts
- First Quartile (Q1) = 25th percentile
- Second Quartile (Q2) = 50th percentile = **Median**
- Third Quartile (Q3) = 75th percentile



- Sorted Marks of 20 students: 65,72,78,80,81,83,85,87,88,90,91,92,93,94,95,96,97,98,99,100
- Position of Q1 = $\frac{\text{Percentile}}{100} \times (n + 1) = \frac{25}{100} \times 21 = \frac{525}{100} = 5.25$, So Q1 = Average of 81 and 83 = 82
- Position of Q2 = $\frac{\text{Percentile}}{100} \times (n + 1) = \frac{50}{100} \times 21 = \frac{1050}{100} = 10.5$, So Q2 = Average of 90 and 91 = 90.5
- Position of Q3 = $\frac{\text{Percentile}}{100} \times (n + 1) = \frac{75}{100} \times 21 = \frac{1575}{100} = 15.75$, So Q3 = Average of 95 and 96 = 95.5

Inter Quartile Range (IQR)

- **Inter Quartile Range (IQR)** = $Q3 - Q1$ = Middle 50% of the data
- In the given example: $IQR = Q3 - Q1 = 95.5 - 82 = 13.5$
- Handles outliers better than range, since the extreme values at both the ends are ignored in IQR
- Since it uses percentiles rather than actual values, it is less affected by **skewed data** (See **Skewness**)
- **Outliers:** Data points that are significantly outside of the typical range of values
 - Lower bound: $Q1 - (1.5 * IQR) = 82 - (1.5 * 13.5) = 61.75$
 - Upper bound: $Q3 + (1.5 * IQR) = 95.5 + (1.5 * 13.5) = 102.25$
 - Points below the *lower bound* or above the *upper bound* are *outliers*
 - In our example, there are no such points, so we do not have any outliers

Outlier Example

- Commute times for 14 randomly selected adults in minutes: 16, 8, 35, 17, 13, 15, 15, 5, 16, 25, 20, 20, 12, 10
- Find outliers and draw a box plot
- Solution: First sort them: 5, 8, 10, 12, 13, 15, 15, 16, 16, 17, 20, 20, 25, 35
- Create a 5-number summary: Minimum, Q1, Q2, Q3, Maximum = 5, 12, 15.5, 20, and 35
- Outlier
 - First calculate $1.5 * IQR = 1.5 \times (20 - 12) = 1.5 \times 8 = 12$
 - Outliers calculation: $Q1 - 12 = 12 - 12 = 0$ and $Q3 + 12 = 20 + 12 = 32$
 - So, outliers = Commute time < 0 or > 32
- Boxplot: Draw a vertical line between 5 and 35; Draw a box with 12 and 20; Draw a median line at 15.5, Show outlier points (See next slide)

Outlier Code

- `import matplotlib.pyplot as plt`
- `import seaborn as sns`

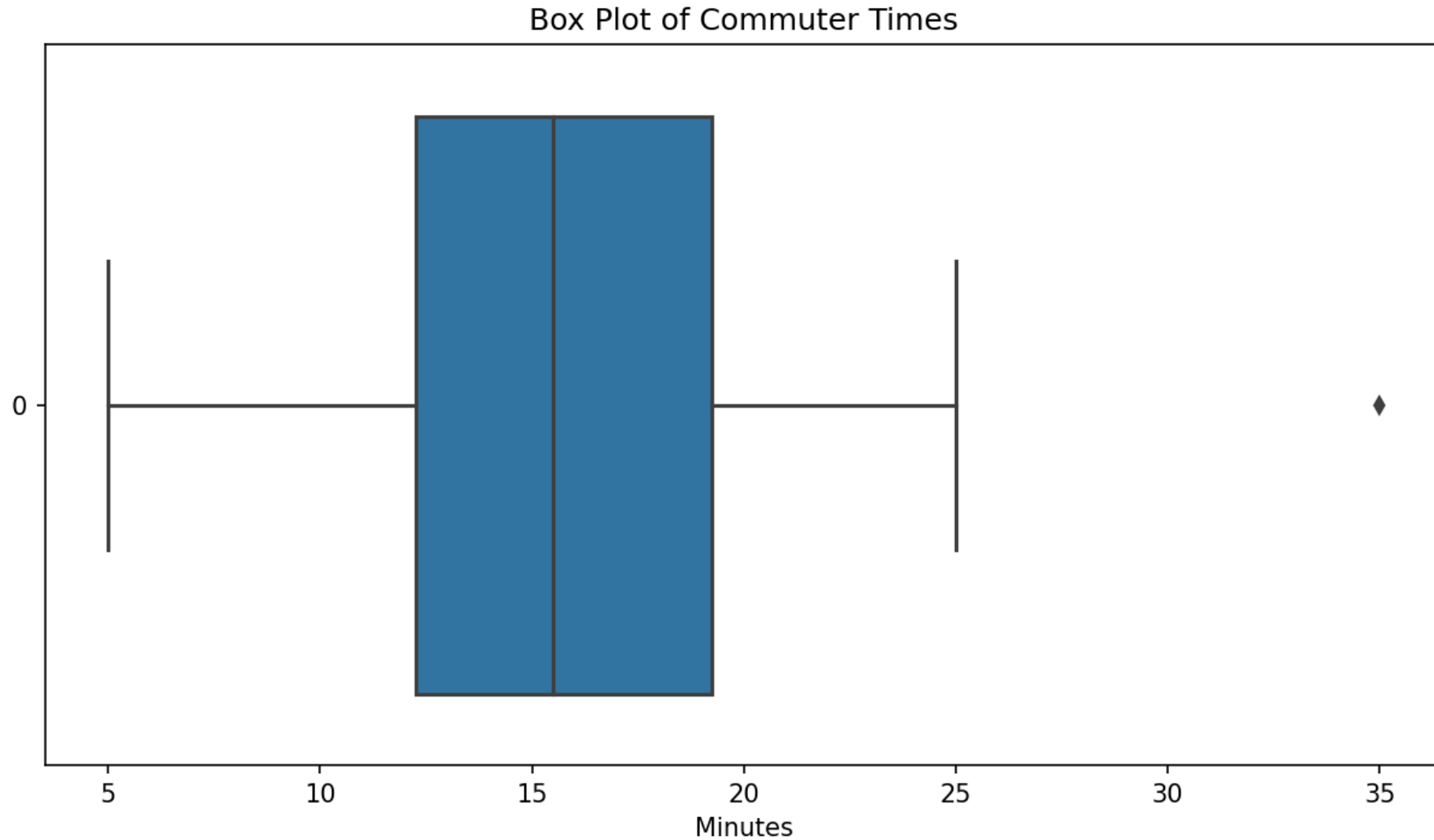
- `# Data`
- `commuter_times = [16, 8, 35, 17, 13, 15, 15, 5, 16, 25, 20, 20, 12, 10]`

- `# Create the box plot`
- `plt.figure(figsize=(10, 6))`
- `sns.boxplot(data=commuter_times, orient='h')`

- `# Add titles and labels`
- `plt.title('Box Plot of Commuter Times')`
- `plt.xlabel('Minutes')`

- `# Show the plot`
- `plt.show()`

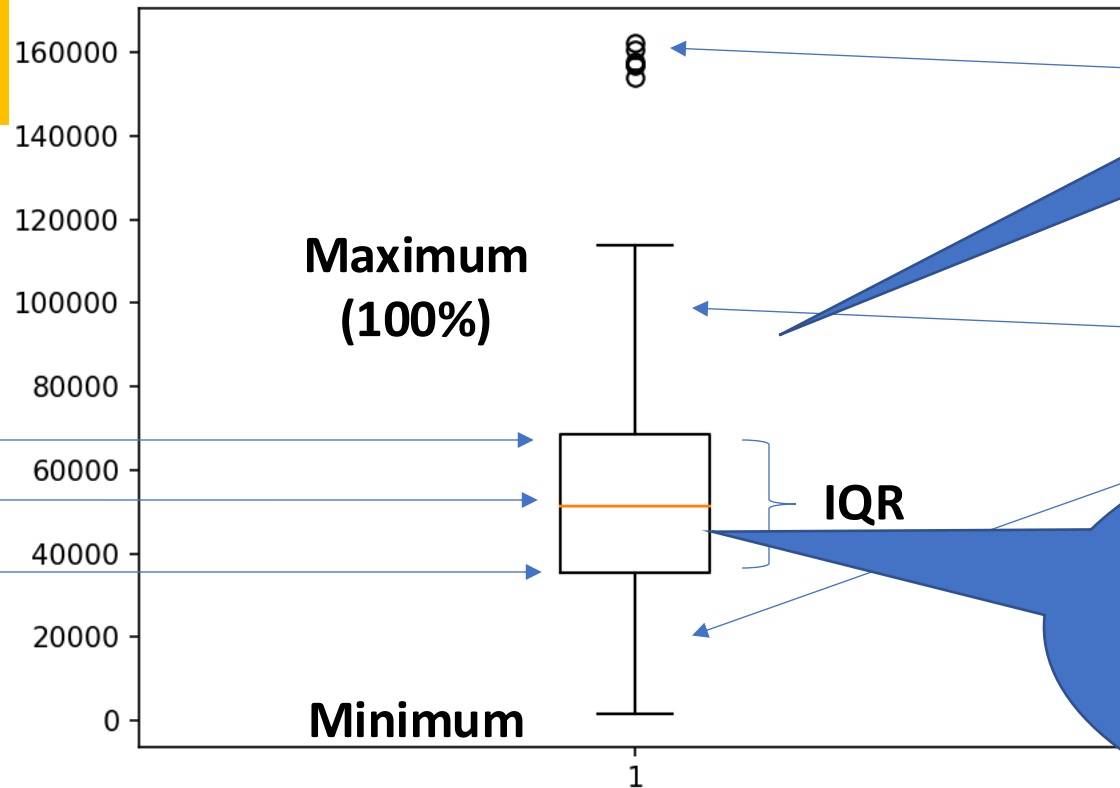
Resulting Boxplot



IQR and Box Plots for Outlier Detection

Outlier: Any data point
below $Q1 - 1.5 * IQR$
or
above $Q3 + 1.5 * IQR$

Q3 (75%)
Q2 (Median) (50%)
Q1 (25%)



Outliers

Whiskers

Box represents middle 50% of the data points.

Short vertical box: Data is condensed around the median,
Long box: Data points are not close to the median

Maximum length of the whisker is 1.5 times that of IQR. Points beyond it are outliers.

Outlier Detection Methods

- **IQR method:** Data points below lower bound or above upper bound
- **Z-Score method:** Data points with Z-score below -3 or above +3 (To be discussed later)
- Code example:
 - C:\code\Data Analytics\outlier_analysis_iqr_z-score_simple.py
 - C:\code\Data Analytics\outlier_analysis_iqr_z-score.py (Different syntax)
- **Assignment:** Do outlier analysis using IQR method for males and females separately for a dataset weight-height.csv
- Enhancement: Use scatterplot to show outliers:
outlier_analaysis_iqr_height_weight.py

Variance

- **Variance** = How far are data points from their mean?
- High variance = Data is more scattered from its mean
- Low variance = Data is less scattered from mean
- Variance calculation:
 1. Find the mean of the given data set
 2. Subtract the mean from each value and square the result
 3. The average of these squared values is the variance

Variance Formula

- **Population variance** (Whole data)

- $\sigma^2 = \sum_{i=1}^N \frac{(x_i - \mu)^2}{N}$

- x_i = Data points
- μ = Population mean
- N = Number of observations (Population size)

- **Sample variance** (Selected data)

- $S^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}$

- x_i = Data points
- $\bar{x}$ = Sample mean
- n = Number of observations (Sample size)

Variance Example

- Consider data values 5, 7, 10, 12, and 15
- Step 1: Mean (μ) : $\frac{(5 + 7 + 10 + 12 + 15)}{5} = 9.8$
- Step 2: Subtract mean from each number and square the result
 - $(5 - 9.8)^2 = 20.24$
 - $(7 - 9.8)^2 = 6.76$
 - $(10 - 9.8)^2 = 0.04$
 - $(12 - 9.8)^2 = 5.44$
 - $(15 - 9.8)^2 = 27.04$
- Step 3: Add up the squared differences and divide by the total number of numbers: $\frac{(20.24 + 6.76 + 0.04 + 5.44 + 27.04)}{5} = 11.12 = \text{Variance } (\sigma^2)$

Why are the Differences Squared?

- If we do not do this, then the positive and negative differences may cancel each other and we may not be able to find how the data points are dispersed from the mean
- Example: Consider marks of 5 students: 55, 60, 65, 54, 64
- Mean: 59.6, Variance: 20.24
- But if we do not square the differences, what will happen?

Marks	Marks - Mean	(Marks - Mean) Squared
55	-4.6	21.16
60	0.4	0.16
65	5.4	29.16
54	-5.6	31.36
64	4.4	19.36
Mean	(Marks - Mean) / 5	((Marks - Mean) Squared) / 5
59.6	0.00	20.24

Variance

Limitation of Variance: How to Quantify?

- Quantifying/measuring variance is tough
- Example: Consider marks of 10 students: 32, 39, 44, 56, 56, 63, 71, 72, 81, 89
- Mean: 60.3, Variance: 306.81
- Variance *seems to be high*, but it does not convey any meaningful information beyond this
- This is a limitation of variance

Standard Deviation

- **Standard deviation** = Square root of variance
- The scale of variance is not to the scale of the original data
- But the scale of standard deviation is the same as scale of the original data
- Variance: Difficult to interpret, Standard deviation: Easy to interpret
- Formula: $\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$

Standard Deviation Example

- Marks of 10 students: 76, 89, 53, 75, 80, 67, 61, 59, 71, 60

- **Mean (μ):**
$$\frac{76 + 89 + 53 + 75 + 80 + 67 + 61 + 59 + 71 + 60}{10} = \frac{691}{10} = 69.10$$

Deviation	Deviation Squared
$76 - 69.10 = 6.9$	47.61
$89 - 69.10 = 19.9$	396.01
$53 - 69.10 = -16.1$	259.21
$75 - 69.10 = 5.9$	34.81
$80 - 69.10 = 10.9$	118.81
$67 - 69.10 = -2.1$	4.41
$61 - 69.10 = -8.1$	65.61
$59 - 69.10 = -10.1$	102.01
$71 - 69.10 = 1.9$	3.61
$60 - 69.10 = -9.1$	82.81
Total:	1114.90
Variance (σ^2) =	1114.90 / 10 = 111.49

- **Standard deviation (σ) = $\sqrt{111.49} = 10.56$**

Exercise

- Interpret the data below
- Group A and Group B are two groups of people and we have data about their blood sugar level
- A = 125, 118, 121, 123, 119, 122, 117, 121, 124, 116, 120, 122, 119, 124, 118, 120, 123, 122, 118, 121
- B = 150, 90, 115, 145, 118, 112, 135, 121, 117, 122, 119, 128, 115, 145, 120, 90, 155, 135, 122, 115

Standard Deviation: True Indicator of Homogeneous/Heterogeneous Data

- Consider cholesterol levels of two groups of people:
- A = 125, 118, 121, 123, 119, 122, 117, 121, 124, 116, 120, 122, 119, 124, 118, 120, 123, 122, 118, 121
- Mean = 120 mg/dL, SD = 5 mg/dL
- Most close to the mean (Fit)
- B = 150, 90, 115, 145, 118, 112, 135, 121, 117, 122, 119, 128, 115, 145, 120, 90, 155, 135, 122, 115
- Mean = 120 mg/dL, SD = 30 mg/dL
- Some close to the mean (Fit), some away from the mean (Unfit)

Interpreting Standard Deviation

- Again consider A = 125, 118, 121, 123, 119, 122, 117, 121, 124, 116, 120, 122, 119, 124, 118, 120, 123, 122, 118, 121
- Mean = 120 mg/dL, SD = 5 mg/dL
- How to calculate *how many standard deviations away* is an observation from the mean?
- Formula: $\frac{\text{Our data point} - \text{Mean}}{\text{Standard Deviation}}$
- Person 1: $\left(\frac{125 - 120}{5}\right) = +1$ standard deviation away from the mean
- Person 2 $\left(\frac{118 - 120}{5}\right) = -0.40$ standard deviations away from the mean

Coefficient of Variation (CV)

- **Coefficient of variation (CV):** Size of standard deviation, related to mean
- Standardized, Unitless
- SD: Expressed in the original scale/units of data (Absolute measure)
- CV: Expressed in percentage, so not dependent on the units of data (Relative measure)
- High CV = Large SD compared to mean ... If it is 100%, it means SD = mean
- $CV = \frac{SD}{Mean} \times 100 \%$
- Example: A pizza restaurant measures its delivery time in minutes. The mean delivery time is 20 minutes and the standard deviation is 5 minutes.
- $CV = \frac{SD}{Mean} \times 100 \% = \frac{5}{20} \times 100 \% = 25\%$

Coefficient of Variation (CV)

- Now suppose a second pizza restaurant has a mean delivery time of 15 minutes and the standard deviation is 10 minutes. Calculate the coefficient of variation and compare with the first one.
- $CV = \frac{SD}{Mean} \times 100 \% = \frac{10}{15} \times 100 \% = 66\%$
- Calculation for the first pizza restaurant was:
- $CV = \frac{SD}{Mean} \times 100 \% = \frac{5}{20} \times 100 \% = 25\%$
- What will be our interpretation?

Making Investment Decisions

- Option A: Average return: 20%, SD: 15%
- Option B: Average return: 6%, SD: 2%
- Option A: $CV = \frac{SD}{Mean} \times 100 \% = \frac{15}{20} \times 100 \% = 75\%$
- Option B: $CV = \frac{SD}{Mean} \times 100 \% = \frac{2}{6} \times 100 \% = 33\%$
- Conclusion: Is Option B better?

Exercise: Calculate CV

- Consider equivalent temperatures

Fahrenheit	Celsius
41	5.00
46	7.88
52	11.11
57	13.89
59	15.00
60	15.56
62	16.67

Solution: Calculate CV

- Consider equivalent temperatures

Fahrenheit	Celsius
41	5.00
46	7.88
52	11.11
57	13.89
59	15.00
60	15.56
62	16.67
SD = 7.28	SD = 4.04
Mean = 53.86	Mean = 12.14
CV = 13.51%	CV = 33.30%

For equal temperatures,
how can CV be different?

Answer: Temperatures use
interval scale

Ideally, we should use CV
only for **ratio** scale
(variables that have an
absolute zero)

Skewness

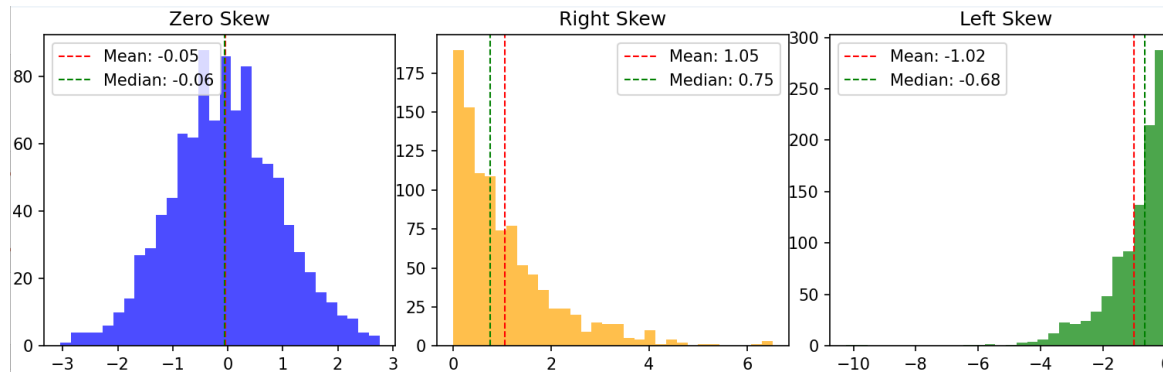
Skewness can be Confusing!

- **Skew**: Represents the asymmetry of a distribution around its mean
- **Skew = Where is the *skew*, not where is the *data***

Skew	Tail	Most Data	Interpretation
Right (Positive)	On the Right	On the Left	Most values are low
Left (Negative)	On the Left	On the Right	Most values are high

- Students appearing in an Examination
- **No-skew**: Student marks are evenly distributed
- **Right-skew**: Most students get low marks => Long tail to the right (higher values)
- **Left-skew**: Most students get high marks => Long tail to the left (lower values)

Skew



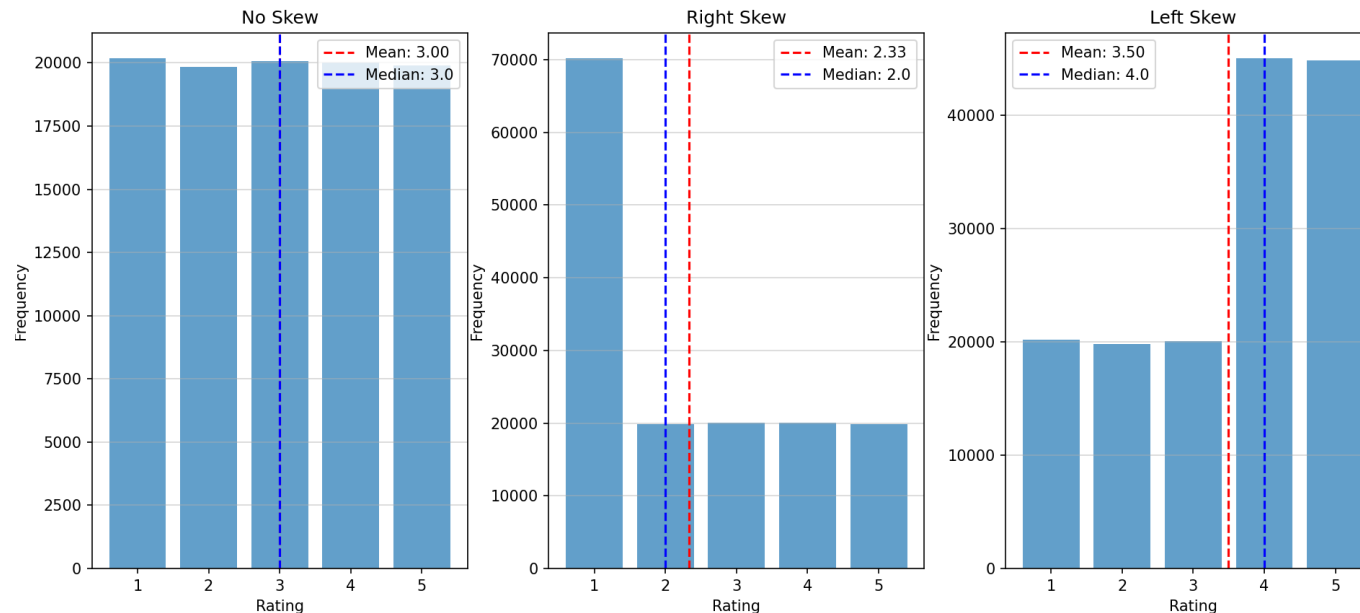
C:\code\Data Analytics\skew-visualize.py

- **Zero-skew (No-skew):** Mean = Median

Characteristic	Right (Positive) Skew	Left (Negative) Skew
Key feature	Most values are on the lower side , a few on are the higher side	Most values are on the higher side , a few are on the lower side
Statistics	Mean > Median	Mean < Median
Mean	Skewed to the right	Skewed to the left
Tail	Comes later - On the right (Represents high values)	Comes first - On the left (Represents low values)
More data points	On the left	On the right
Outliers	On the right	On the left
Ratings Example	Most people give ratings of 1 or 2 out of 5	Most people give ratings of 4 or 5 out of 5
Example	Incomes of people	Ages at the time of death
Why?	Most people have low incomes, Very few have very high income	Most people would die in older age, Very few children die

Simple Ratings Example (C:\code\Data Analytics\Skew-dummy.py)

- Consider 1 lakh people giving ratings on a scale of 1-5
- **No-skew**: All ratings occur almost with the same frequency
- **Right-skew**: Most people give low ratings (1/2)
- **Left-skew**: Most people give high ratings (4/5)



Skew Real-Life Examples

- **Right** Skew (More data points are on the **left**)
- C:\code\Data Analytics\skewness-airlines-distance.py
- **Left** Skew (More data points are on the **right**)
- C:\code\Data Analytics\skewness-airlines-onboard-service.py (Note: The dataset is different here)

Skewness Formulae

- **Pearson's First Coefficient of Skewness** (Mode Skewness): $(\text{Mean} - \text{Median}) / \text{Standard Deviation}$
- **Another variation:** $3 (\text{Mean} - \text{Median}) / \text{Standard Deviation}$
- **Fisher-Pearson Coefficient of Skewness** (Moment Skewness): $(\sum (x_i - \text{Mean})^3 / n) / [(\sum (x_i - \text{Mean})^2 / n)^{(3/2)}]$
- **Quantile-Based Skewness:** $(Q3 + Q1 - 2 * \text{Median}) / (Q3 - Q1)$

Pearson's Coefficient of Skewness

- Formula: $Skewness = \frac{3 (Mean - Median)}{Standard\ Deviation}$
- Consider daily commute time for a set of people in minutes:
20,22,25,30,35,40,45,50,60,70
- Mean = $397/10 = 39.70$
- Median: 37.5
- SD = 15.80
- $Skewness = \frac{3 (Mean - Median)}{Standard\ Deviation} = \frac{3 (39.70 - 37.70)}{15.80} = 0.37$
- Interpretation: Right/positive skew ... Most commute times are less than the mean, but a few outliers have increased the mean value

Covariance

Covariance

- **Covariance:** Quantifies how changes in one variable relate to changes in another
- Scale:
 - Covariance: Units of what we are measuring (difficult to interpret)
 - Correlation: -1 and +1 (easy)

- Example:

Area of the House	Price of the House
1200	75 lakhs
1300	90 lakhs
1500	1.10 crores

- As area increases, price increases: But by how much? ... Covariance

Covariance Example

Height	Weight
65	68
67	69
68	70
66	69
64	65

- $\text{Cov}(x,y) = \sum_{i=1}^n \frac{(x_i - \bar{x})(y_i - \bar{y})}{n-1}$
- Mean height ($\bar{x}$) = $\frac{65+67+68+66+64}{5} = 66$
- Mean weight ($\bar{y}$) = $\frac{68+69+70+69+65}{5} = 68.2$
- For Heights: $x_i - \bar{x} = [65-66, 67-66, 68-66, 66-66, 64-66] = [-1, 1, 2, 0, -2]$
- For Weights: $y_i - \bar{y} = [68-68.2, 69-68.2, 70-68.2, 69-68.2, 65-68.2] = [-0.2, 0.8, 1.8, 0.8, -3.2]$
- $(x_i - \bar{x})(y_i - \bar{y}) = (-1 \times -0.2) + (1 \times 0.8) + (2 \times 1.8) + (0 \times 0.8) + (-2 \times -3.2) = 11$
- $\text{Cov}(x,y) = \sum_{i=1}^n \frac{(x_i - \bar{x})(y_i - \bar{y})}{n-1} = \frac{11}{4} = 2.75$

Covariance and Variance

- As we know, $\text{Cov}(x, y) = \sum_{i=1}^n \frac{(x_i - \bar{x})(y_i - \bar{y})}{n-1}$
- What if we use x instead of y ?
- $\text{Cov}(x, x) = \sum_{i=1}^n \frac{(x_i - \bar{x})(x_i - \bar{x})}{n-1}$
- Remember: $\text{Variance}(x) = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1} = \sum_{i=1}^n \frac{(x_i - \bar{x})(x_i - \bar{x})}{n-1}$
- So: $\text{Var}(x) = \text{Cov}(x, x)$

Covariance Example

- C:\code\Data Analytics\correlation_covariance_height_weight.py

```
Overall Covariance between Height and Weight:  
      Height      Weight  
Height  14.803473  114.242656  
Weight 114.242656 1030.951855
```

```
Overall Correlation between Height and Weight:  
      Height      Weight  
Height  1.000000  0.924756  
Weight  0.924756  1.000000
```

- Diagonal elements: Variance of the variable (Height: 14, Weight: 1030)
- Off-diagonal elements: Covariance between the two variables (114)
 - Since $114 > 0$: As height increases/decreases, so does weight
 - Covariance = 114: No real interpretation on its own
 - But Correlation = 0.92: Very easy to interpret

Covariance Interpretation

- Compare covariance for three cases: (1) overall (2) males (3) females

Overall covariance = 114, For males = 48, For females = 43

Overall Covariance between Height and Weight:

	Height	Weight
Height	14.803473	114.242656
Weight	114.242656	1030.951855

Male Covariance between Height and Weight:

	Height	Weight
Height	8.198843	48.879649
Weight	48.879649	391.294074

Female Covariance between Height and Weight:

	Height	Weight
Height	7.269947	43.576404
Weight	43.576404	361.854281

Other statistics (Calculated separately):

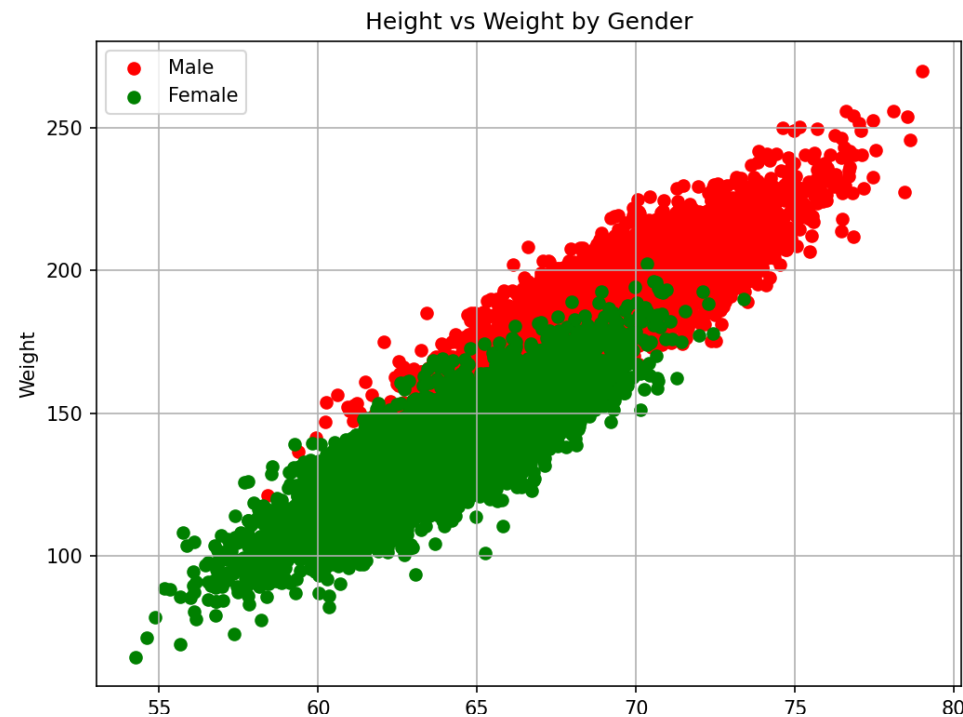
male records: 5001, average height: 69, average weight: 187

female records: 5001, average height: 63, average weight: 135

- (1) Heights and weights are positively related in all three cases
- (2) Gender difference: Males are consistently taller and have more weight
- (3) So, although *within group* covariances are low, difference in means contributes to a higher *overall covariance*
- (4) That is, higher covariance *pulls* the lower covariance to a higher value (See next slide)

Covariance Interpretation

- Plotting all data points: C:\code\Data Analytics\scatter_plot_heights_weights.py
- Higher covariance in male records *pulls* lower covariance of females records to a higher side



Correlation

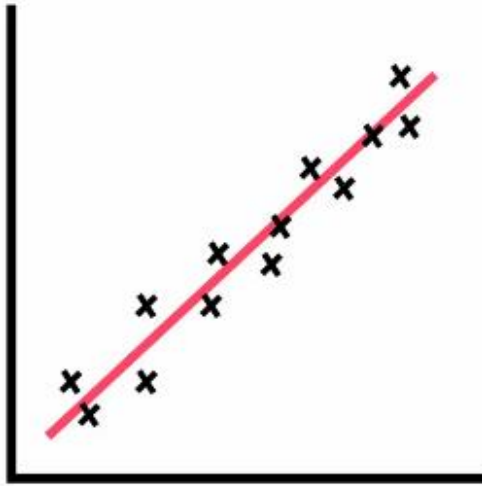
Correlation

- **Correlation:** Are two numeric variables related?
- Examples: Exercise and Health, Study and Marks, Experience and Salary
- Correlation $(x,y) = \frac{Cov(x,y)}{\sigma_x \sigma_y}$
- Measured using **correlation coefficient (r)**
 - **Pearson Correlation Coefficient:** Measures linear relationships between continuous variables
 - **Spearman Rank Correlation Coefficient:** Measures relationships, even if they are not strictly linear

Pearson Correlation Coefficient Interpretation

- Range: -1 to +1
- Positive correlation: Variables move up together
 - Example: Correlation of 0.80 between Hours spent studying and test scores
- Negative correlation: As one variable moves up, the other moves down
 - Example: Correlation of -0.70 between Hours spent watching TV and physical fitness
- Zero correlation: Variables are unrelated
 - Example: Correlation of 0.02 between Shoe size and IQ score

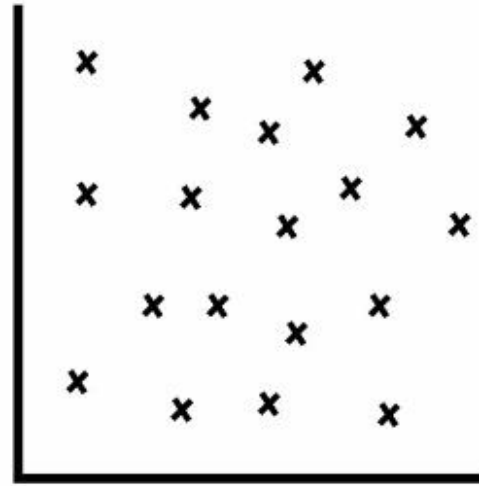
Positive, Negative, No Correlation



Positive
Correlation



Negative
Correlation



No
Correlation

Pearson Correlation Coefficient for Earlier Example

- $\text{Corr}(x,y) = \frac{\text{Cov}(x,y)}{\sigma_x \sigma_y}$
- $\text{Corr}(x,y) = \frac{2.75}{1.58 \times 1.92}$
- $\text{Corr}(x,y) = \frac{2.75}{1.58 \times 1.92} = 0.90$

Height	Weight
65	68
67	69
68	70
66	69
64	65

Key Point about Correlation

- Find correlation between related variables (e.g. *hours of study* and *marks obtained*, but not *hours of study* and *time needed to reach C-DAC*)
- See C:\code\Data Analytics\correlation-outliers-bad-example.py
- Better example: correlation_better_example.py
- Good example: correlation-positive-currency-rates.py

Correlation Examples

- C:\code\Data Analytics\correlation-2.py
- C:\code\Data Analytics\correlation_poor.py
- Assignment: Find the correlation between *pace* and *official time* columns of the combined marathon datasets (C:\code\Data Analytics\correlation_good_marathon.py)

Covariance Example

- C:\code\Data Analytics\correlation-covariance-crypto-gold.py

- Interpretation: Correlation

Correlation Matrix:

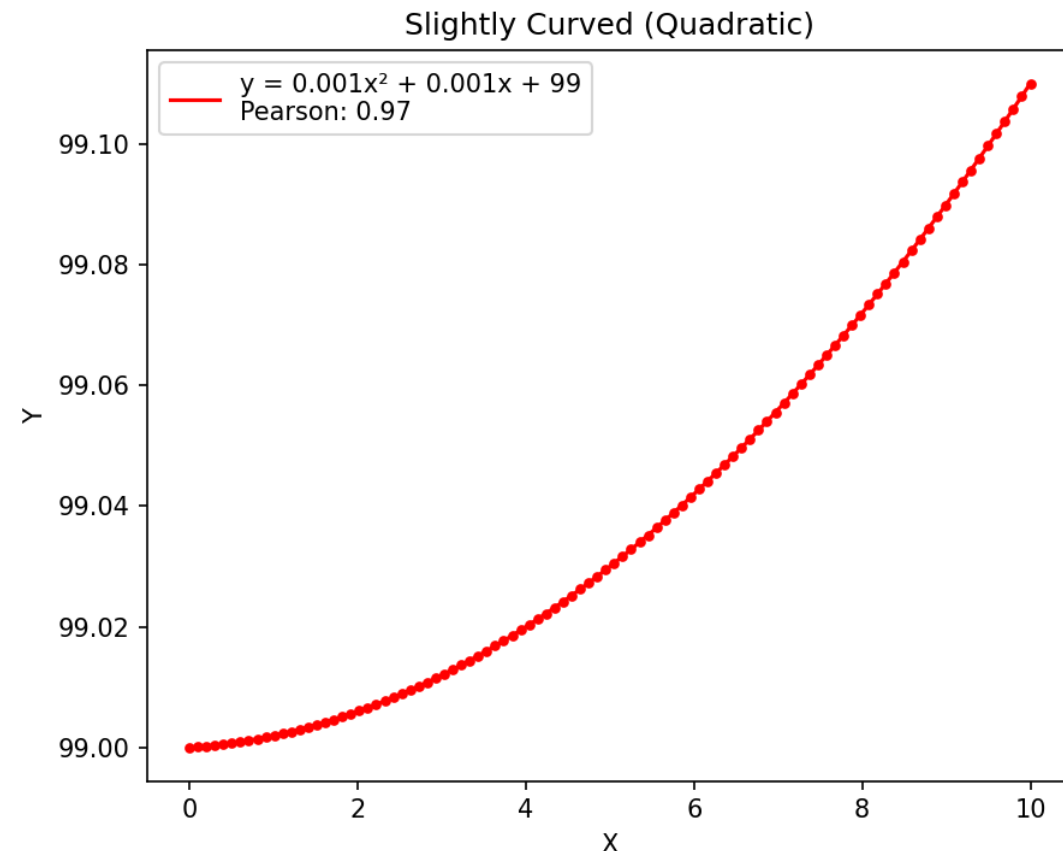
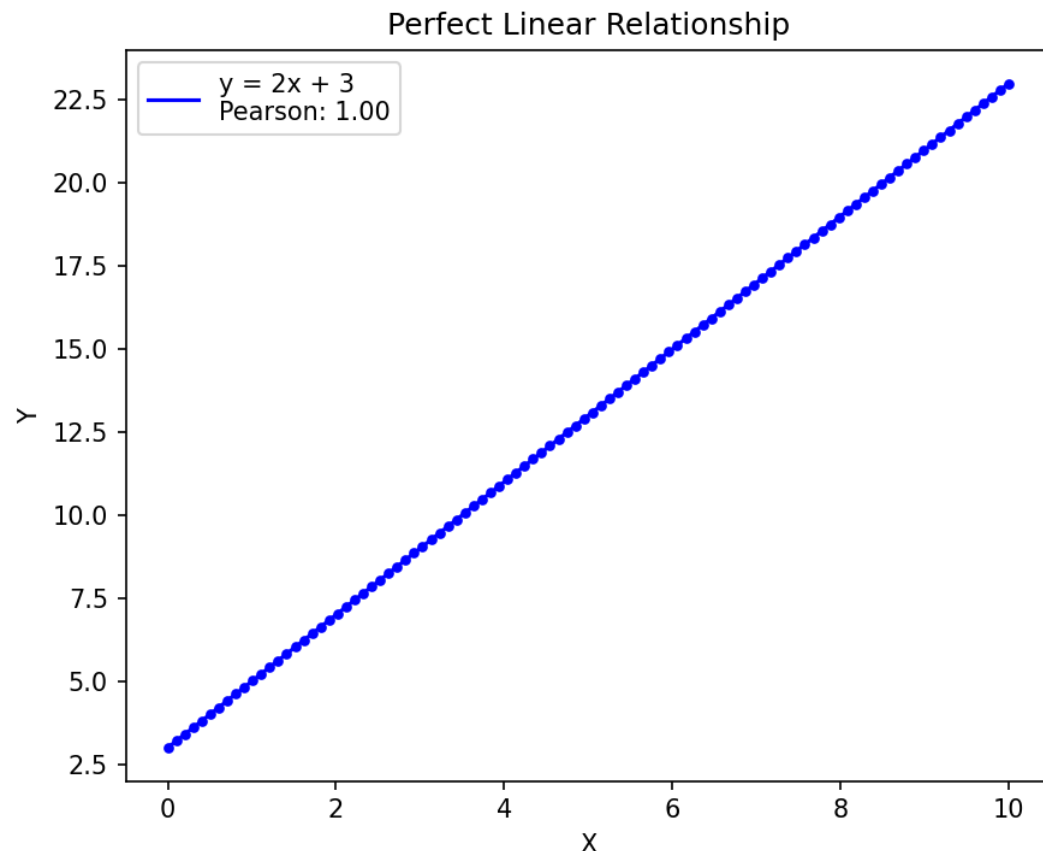
	BITCOIN	Gold	S&P500
BITCOIN	1.000000	0.594622	0.862225
Gold	0.594622	1.000000	0.445658
S&P500	0.862225	0.445658	1.000000

- Interpretation: Covariance

	BITCOIN	Gold	S&P500
BITCOIN	213488571.4899	2124359.2051	11870614.3705
Gold	2124359.2051	59786.0101	102675.5060
S&P500	11870614.3705	102675.5060	887830.9549

Why Spearman Rank Correlation?

- Pearson correlation coefficient: Works well when data is linear, but not well when the data is not linear



Spearman Rank Correlation

- $r_s = \frac{\text{Cov}(R(x), R(y))}{\sigma(R(x)) \sigma(R(y))}$... $R(x)$ and $R(y)$ specify the ranks of x and y
- More popular formula: $r_s = 1 - \frac{6 \sum d^2}{(n^2 - 1)}$
- d^2 = Square of differences in the ranks of the two coordinates for each point (x, y)
- n = Number of data points

Spearman Rank Correlation Coefficient for Earlier Example

Height	Weight
65	68
67	69
68	70
66	69
64	65

- $r_s = \frac{Cov(x,y)}{\sigma_x \sigma_y}$
- More popular formula: $r_s = 1 - \frac{6 \sum d^2}{(n^2 - 1)}$
- d^2 = Square of differences in the ranks of the two coordinates for each point (x,y)
- n = Number of data points

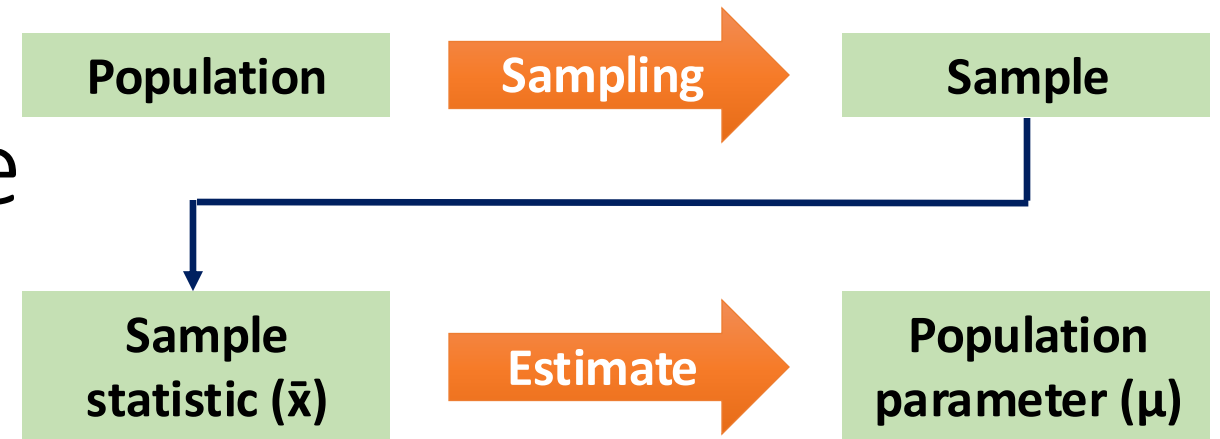
$$r_s = 1 - \frac{6 \sum d^2}{(n^2 - 1)} = 1 - \frac{6 \times 0.50}{(5^2 - 1)}$$

$$r_s = 1 - \frac{3}{24} = 1 - 0.125 = 0.87$$

x_i	R_x	y_i	R_y	$d_i = R_x - R_y$	d_i^2
65	2	68	2	0	0
67	4	69	3.5	0.5	0.25
68	5	70	5	0	0
66	3	69	3.5	-0.5	0.25
64	1	65	1	0	0

Population and Sample, Probability Theory

Population and Sample



- 10000 students in a university
- Question: What is the average student height?
 - Difficult: Measure the actual height of the **population**, i.e. all 10000 students to calculate the average (**parameter**)
 - Easier: Take a **sample** of 500 students, calculate the average and use it to estimate the population average (**statistic**)

Population Parameter	Symbol	Sample Statistic	Symbol
Mean	μ (Mu)	Sample Mean	$\bar{x}$ (x bar)
Standard Deviation	σ (Sigma)	Sample Standard Deviation	s
Proportion	π (Pi)	Sample Proportion	$\hat{p}$ (p-hat)

Sampling Types

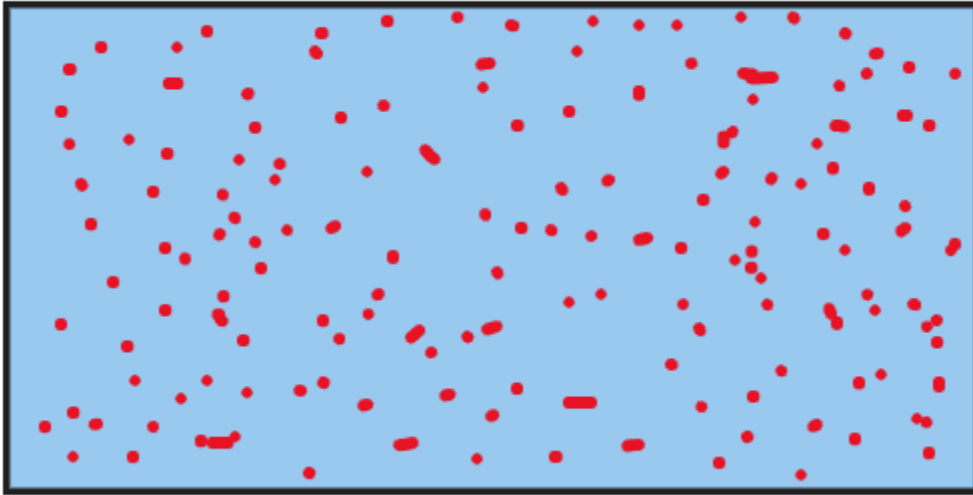
Univariate Sampling	Bivariate Sampling
Analysis of a single variable	Analysis of two variables to find relationship between them
Analyzing the ages of individuals in a population Examining the heights of a sample of students Studying the test scores of students in a class	Amount of time spent studying (variable 1) and the Test scores achieved (variable 2) Education (variable 1) and the Income level (variable 2)
Descriptive statistics such as mean, mode, median, variance, standard deviation	Correlation, Regression
Histogram, Box plot	Scatter plot, Heatmap

Resampling

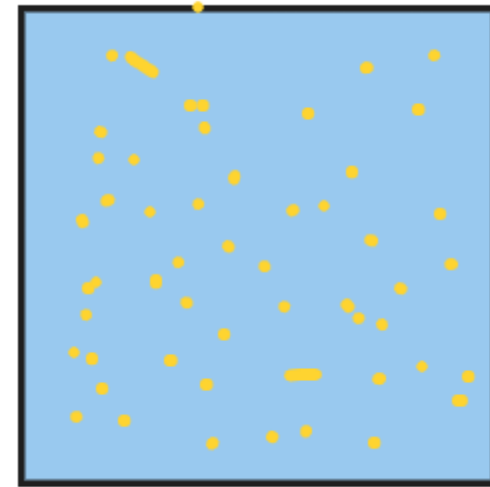
- **Resampling:** Modifying a dataset by changing its *size* or *composition*
- How? **Bootstrapping** or **Cross-validation**
- **Bootstrapping:** Generating multiple new samples (called **bootstrap samples**) from the original dataset by randomly selecting data points with replacement – Useful in estimating population parameters, regression analysis
- **Cross-validation:** Dividing the dataset into a **training set** and a **testing set**, and iteratively training and evaluating the model on different combinations of these subsets – Useful in **machine learning** model evaluation (See next slide)

Sampling and Population

- **Population (N)** = Complete data



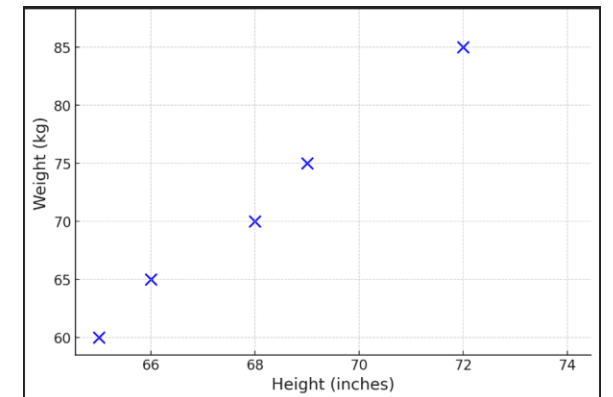
- **Sample (n)** = Subset of data



- **Estimation:** From sample mean $\rightarrow$ Population mean, i.e. $\bar{x} \rightarrow \mu$
- **Point estimate:** Exact estimate, **Interval estimate:** A range of estimates (Called **Confidence interval**)

Cross-validation

- Problem: Not sure which data to use for Training and which for Testing
- Solution: **Cross-validation**
- How? Use all the points for both Training and Testing iteratively (i.e. in steps)
- Suppose we have weight and height data for six people
- Question: How to decide what data to use for Training and what data to use for Testing?
- Bad idea: Use all the data for both
- Why? May cause **overfitting** (See next slide)



Overfitting and Holdout Evaluation

- A student who memorizes the answers to a specific set of questions for an exam but fails to answer questions on a different test
- **Memorizing** versus **Learning**
- **Overfitting**: A machine learning model performs exceptionally well on the training data but fails on new data
- Overfit models *memorize* the training data rather than *learning* the underlying patterns
- Hence, we must not use the entire data for training, but split data into something like 80% training and 20% testing
- **Holdout evaluation** : Use the testing set to see how well the model performs

K-Fold Cross-Validation

- **K-fold cross-validation:** Divide data into k groups (**folds**) and perform training and testing k times, once with each fold as the test set
- Solution to overfitting

Iteration	Training sets	Test set
1	1, 2	3
2	2, 3	1
3	3, 1	2

- Select the one which gives the best results

Sampling Techniques

- **Probability sampling:** Random selection
 - Every member of the population has a chance of being selected
 - Most appropriate
 - Mainly used in quantitative research
- **Non-probability sampling:** Non-random selection
 - Not every individual has a chance of being included
 - Easier and cheaper to access, has a higher risk of **sampling bias**
 - Used in qualitative research

Probability Sampling

Type	Brief description	Method	Example
Simple	Every member of the population has an equal chance	Random number generator	Assign # 1 to 1000 to employees and select n random samples
Systematic	Similar but easier	Choose at regular intervals	Pick employee # 7, 17, 27, 37, 47, ...
Stratified	More representative, if we have sub-categories within data	Divide population into strata (sub-groups) and then pick proportionate samples from each	If we have 700 women employees and 300 men employees, pick 70 women and 30 men in the sample
Cluster	Similar, but use entire sub-groups, rather than samples within sub-groups	Divide population into clusters and pick entire clusters as samples	If the offices are in Pune, Mumbai, Delhi, Bangalore, Hyderabad, and Chennai select Pune and Chennai as samples

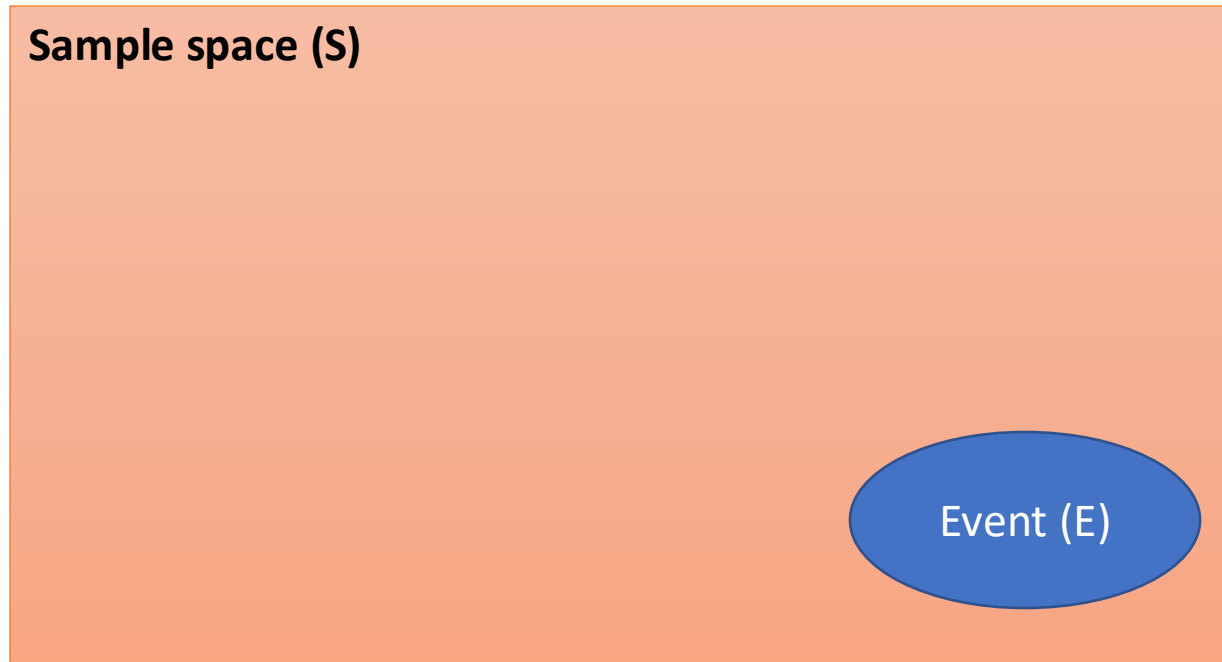
Non-Probability Sampling

Type	Brief description	Method	Example
Convenience	Just pick sample data as per one's convenience	No systematic method	To research opinions about student support services in our university, we ask our fellow students to complete a survey on the topic
Voluntary-response	People voluntarily come forward	Start with the people who are interested – May be biased	Ask employees about problems faced through a third-party survey – unhappy employees will come forward on their own
Purposive	Done for a specific purpose, also called judgment sampling	Reach the right people and then do a detailed study	We want to know the opinions of foreign students at C-DAC, so we specifically talk to them about a variety of topics
Snowball	Recruit more participants via existing participants	Create a snowball of participants	Homeless people survey – Initially difficult to find such people to speak – But they may help us speak with others soon

Sample Space and Events, Probability

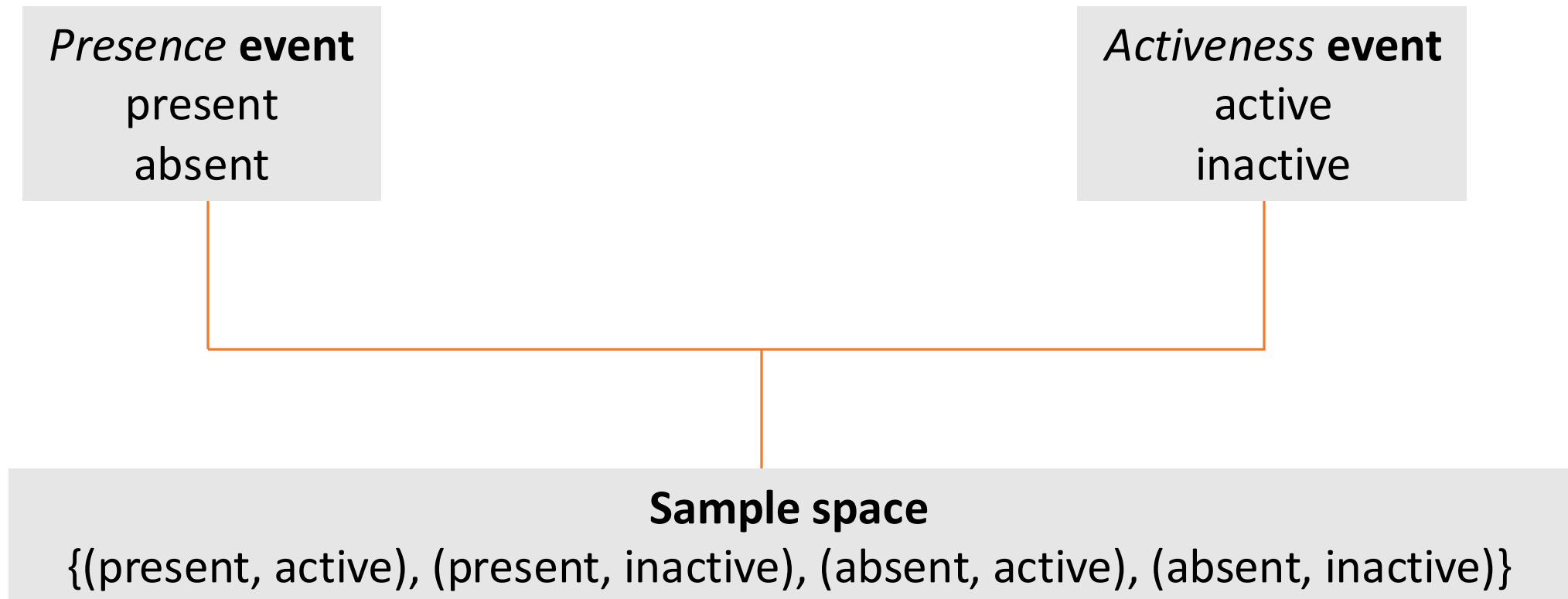
Sample Space and Events

- **Sample space:** All the possible outcomes of an experiment
- **Event:** A subset of the sample space

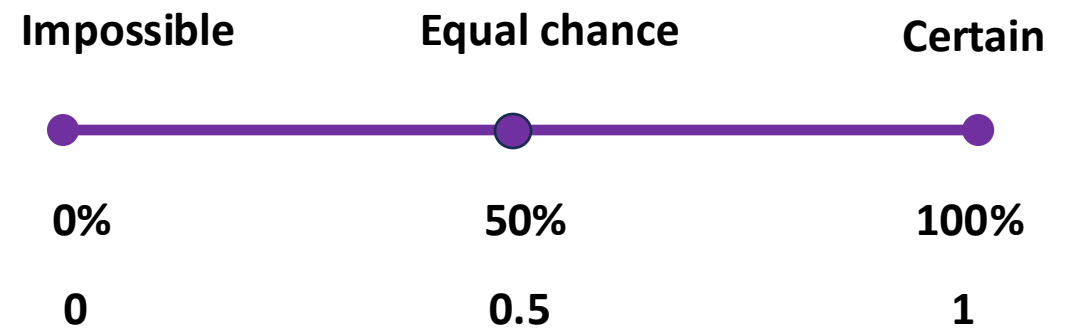


Events and Sample Space: Examples

- Consider students in our class



Probability Definition



- **Probability:** How likely it is that something will occur?
- Written as a number or fraction (e.g. 0.5 or 50%)
- Example:
 - What is the probability that it will rain today?
 - Sample space (S) = {R, NR}
 - $P(R) = 1 / 2 = 0.5$ or 50%
 - $P(NR) = 1 / 2 = 0.5$ or 50%
- But what about rainy season?
 - It should be: $P(R) = 100\%$ and $P(NR) = 0\%$
- **Theoretical probability:** What will happen in theory?
- **Empirical/Practical probability:** What actually happened?

Theoretical probability

Empirical probability

A team winning consecutive matches (Symbols: Win = 1, Loss = 0)

Match	Possible cases	Theoretical probability	Theoretical probability
1	0 1	$\frac{1}{2}$	0.50 = 50%
2	00 01 10 11	$\frac{1}{4}$	0.25 = 25%
3	000 001 010 011 100 101 110 111	$\frac{1}{8}$	0.125 = 12.5%
4	0000 0001 0010 0011 0100 0101 0110 0111 1000 1001 1010 1011 1100 1101 1110 1111	$\frac{1}{16}$	0.0625 = 6.25%

Imagine the team actually winning four consecutive matches

Theoretical probability would diminish to 6.25%

But the team has actually won! So, the empirical probability will be 100%

Marginal, Joint, Conditional Probability

Basic Terms

- **Marginal probability:** Probability of an event irrespective of the outcome of another variable: $P(A)$
- **Joint probability:** Probability of two events occurring **without any interdependence**: $P(A \cap B)$ or $P(A,B)$
- **Conditional probability:** Probability of a second event happening **given that** the first one has happened: $P(B|A) = \frac{P(A,B)}{P(A)}$
- Joint probability: Outcome of B **is not** influenced by A
- Conditional probability: Outcome of B **is** influenced by A

Marginal Probability

- Example: In Pune, 70% people like apple and 30% people like orange
- $P(\text{Apple}) = 70\%$
- $P(\text{Orange}) = 30\%$
- Both are examples of **marginal probability**

Independent Events

- An **independent** series of events occur when the outcome of one event has no effect on the outcome of another
- An example is flipping a fair coin twice
- The chance of getting heads on the second toss is independent of the result of the first toss

Independent Events – Example

- The probability of seeing two heads with two flips of a fair coin is:

$$P(H1 \text{ } H2) = P(H1) \times P(H2)$$

$$= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

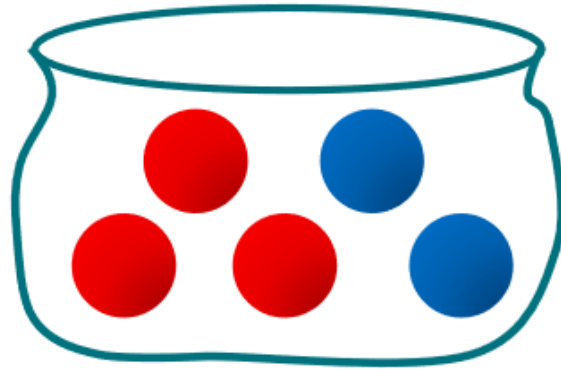
1 st Toss	2 nd Toss
H	H
H	T
T	H
T	T

Dependent Events

- A **dependent** event occurs when the outcome of a first event does affect the probability of a second event
- Example: Draw colored marbles from a bag without replacement

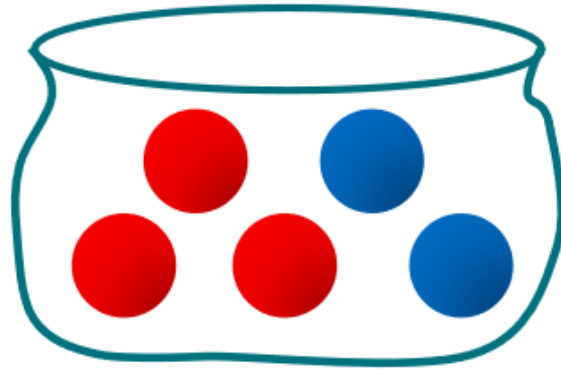
Dependent Events

- Imagine a jar contains 2 blue marbles and 3 red marbles
- If you take two marbles out of the jar, what is the probability that they are both red?



Dependent Events

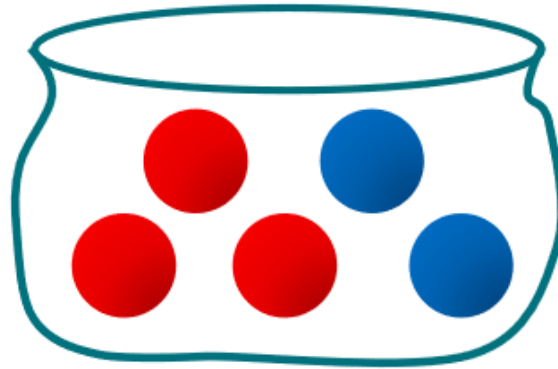
- Here the color of the first marble affects the probability of drawing a 2nd **red** marble



Dependent Events

- The probability of drawing a first **red** marble is easy:

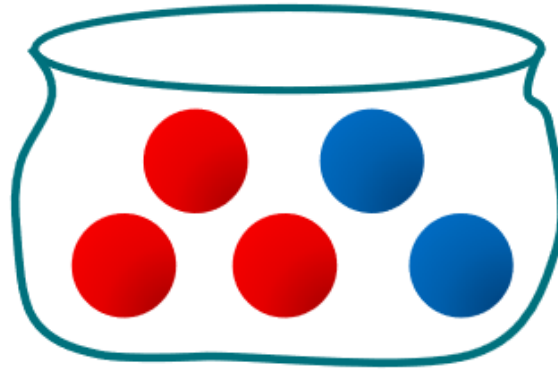
$$P(R1) = \frac{3}{5}$$



Dependent Events

- The probability of drawing a second **red** marble given that the first marble was **red** is written as:

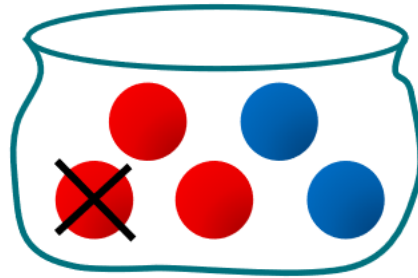
$$P(R2 | R1)$$



Dependent Events

- After removing a **red** marble from the sample set this becomes :

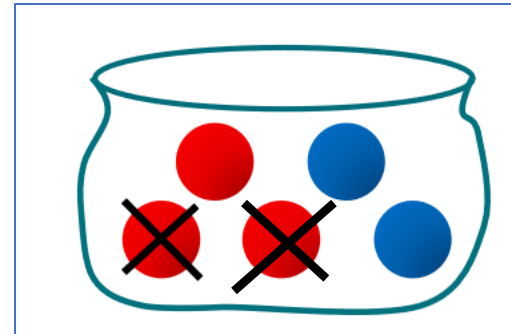
$$P(R2 | R1) = \frac{2}{4}$$



Dependent Events

- So the probability of two **red** marbles is:

$$P(R1 \cap R2) = P(R1) \times P(R2 | R1) = \frac{3}{5} \times \frac{2}{4} = \frac{6}{20} = 0.3$$



Conditional Probability to Bayes' Theorem

- **Bayes' Theorem:** Helps reverse conditional probabilities by incorporating conditional probability

- **Conditional probability**

- $P(B|A) = \frac{P(A,B)}{P(A)}$

- **Bayes' Theorem**

- $P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$

Conditional Probability to Bayes' Theorem

- Conditional Probability: $P(\text{Good Credit Rating} \mid \text{Loan Approved})$
- Bayes' Theorem: $P(\text{Loan Approved} \mid \text{Good Credit Rating})$
- Suppose $A = \text{Good credit rating}$, $B = \text{Loan approved}$
- As we know, Conditional Probability: $P(A \mid B) = \frac{P(A,B)}{P(B)}$
- **$P(\text{Good credit rating} \mid \text{Loan approved}) =$**
$$\frac{P(\text{Good credit rating and Loan approved})}{P(\text{Loan approved})}$$
- Now Bayes' Theorem: $P(B \mid A) = \frac{P(A \mid B) \times P(B)}{P(A)}$
- $P(\text{Loan approved} \mid \text{Good credit rating}) =$
$$\frac{P(\text{Good credit rating} \mid \text{Loan approved}) \times P(\text{Loan approved})}{P(\text{Good credit rating})}$$

Bayes' Theorem Example

- 10% of patients entering a clinic have liver disease
- 5% of patients entering a clinic are alcoholic
- Out of the patients who have liver disease, 7% are alcoholics
- A = Liver disease; So $P(A) = 0.10$
- B = Alcoholic; So $P(B) = 0.05$
- $B|A$ = Patient is alcoholic, given that the patient has liver disease; So $P(B|A) = 0.07$
- Find $P(A|B)$, i.e. probability that the patient has liver disease, given that the patient is alcoholic

Bayes' Theorem Example

- $P(\text{Liver disease}|\text{Alcoholic}) = \frac{P(\text{Alcoholic}|\text{Liver disease}) \times P(\text{Liver disease})}{P(\text{Alcoholic})}$
- i.e. $P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$
- $P(A|B) = \frac{0.07 \times 0.1}{0.05}$
- $P(A|B) = 0.14$
- Conclusion: If the patient is an alcoholic, their chances of having liver disease is 0.14 (14%)

Bayes' Theorem Example

- Rain prediction
- Overall historical probability of rain: $P(R) = 0.30$
- Sky condition
 - $P(\text{Overcast} | \text{Rain}) = 0.8$
 - $P(\text{Clear-sky} | \text{Rain}) = 0.2$
 - $P(\text{Overcast}) = 0.6$
- Find $P(\text{Rain} | \text{Overcast})$, because today it is overcast
- Suppose $A = \text{Rain}$, $B = \text{Overcast sky condition}$

Bayes' Theorem Example

- $P(Rain|Overcast) = \frac{P(Overcast|Rain) \times P(Rain)}{P(Overcast)}$
- i.e. $P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$
- $P(Rain|Overcast) = \frac{0.8 \times 0.3}{0.6}$
- $P(Rain|Overcast) = 0.4$ or 40%

Bayes' Theorem

- Dangerous fires are rare (1%)
- Smoke is quite common due to barbecues (10%)
- 90% dangerous fires cause smoke
- What is the probability that we have a dangerous fire when there is smoke?

Bayes' Theorem

- Dangerous fires are rare (1%), Smoke is quite common due to barbecues (10%), 90% dangerous fires cause smoke
- What is the probability that we have a dangerous fire when there is smoke? Suppose $A = \text{Fire}$, $B = \text{Smoke}$

- $$P(A | B) = \frac{P(B|A) \times P(A)}{P(B)}$$

- $$P(\text{Fire} | \text{Smoke}) = \frac{P(\text{Smoke} | \text{Fire}) \times P(\text{Fire})}{P(\text{Smoke})}$$

- $$P(\text{Fire} | \text{Smoke}) = \frac{90\% \times 1\%}{10\%}$$

- $$P(\text{Fire} | \text{Smoke}) = 9\%$$

Random Variables

Random Variables

- In an experiment, we may be interested in some *function of the outcome*, rather than the *outcome itself*
- Example: In 1000 credit card transactions
 - Our interest: What was the fraud ratio
 - Not of our interest: The outcome of every single transaction
- **Random variables** = These functions on the sample space
- Example: 3 credit card transactions, Y = Random variable = Number of frauds ... Let F = Fraud and N = No fraud
 - $P(Y = 0) = P\{(NNN)\} = \frac{1}{8}$
 - $P(Y = 1) = P\{(NNF), (NFN), (FNN)\} = \frac{3}{8}$
 - $P(Y = 2) = P\{(NFF), (FNF), (FFN)\} = \frac{3}{8}$
 - $P(Y = 3) = P\{((FFF))\} = \frac{1}{8}$

Random Variables: Example

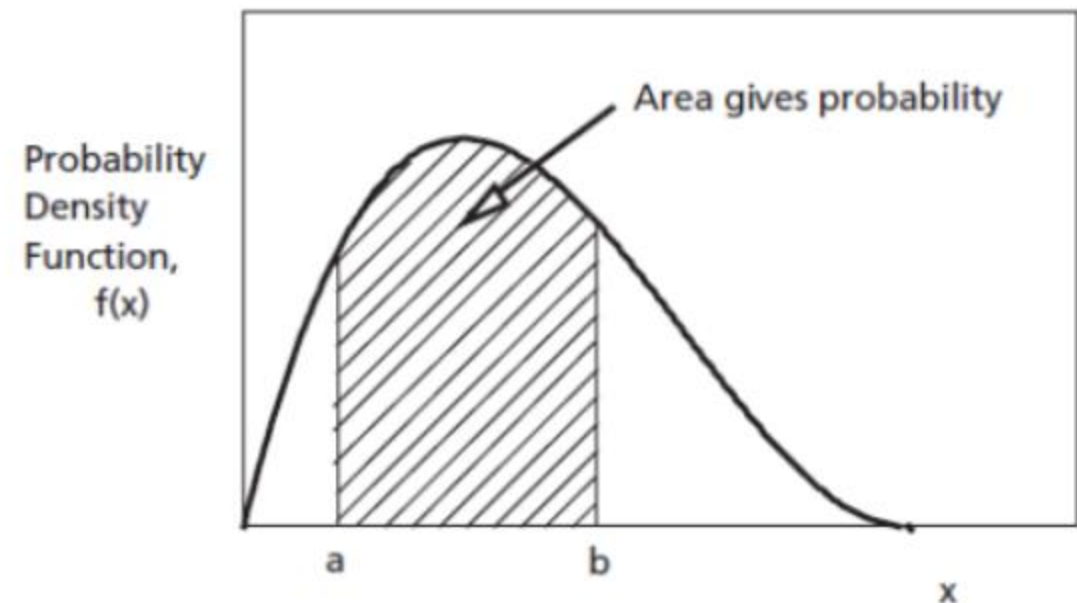
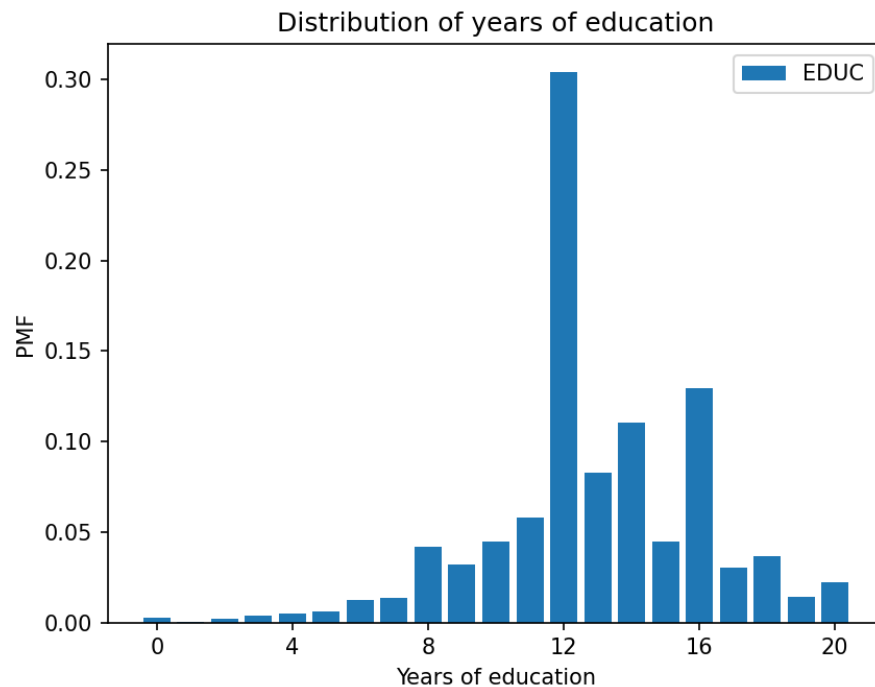
- A life insurance agent has two elderly clients with policy value of Rs 1 crore upon death
- Event Y = Younger of the two dies, $P(Y) = 0.05$
- Event O = Older of the two dies, $P(O) = 0.10$
- Y and O are independent
- X = Random variable representing the amount of money paid out (in crores) based on who dies
- Possible values for X : 0, 1, 2 crore
- Nobody dies: $P(X = 0) = P(Y^c O^c) = 0.95 \times 0.90 = 0.855$ or 85.5%
- One of them dies: $P(X = 1) = P(Y O^c) + P(Y^c O) = (0.05 \times 0.90) + (0.95 \times 0.10) = 0.140$ or 14%
- Both die: $P(X = 2) = P(Y O) = (0.05 \times 0.10) = 0.005$ or 0.5%

Random Variable Types

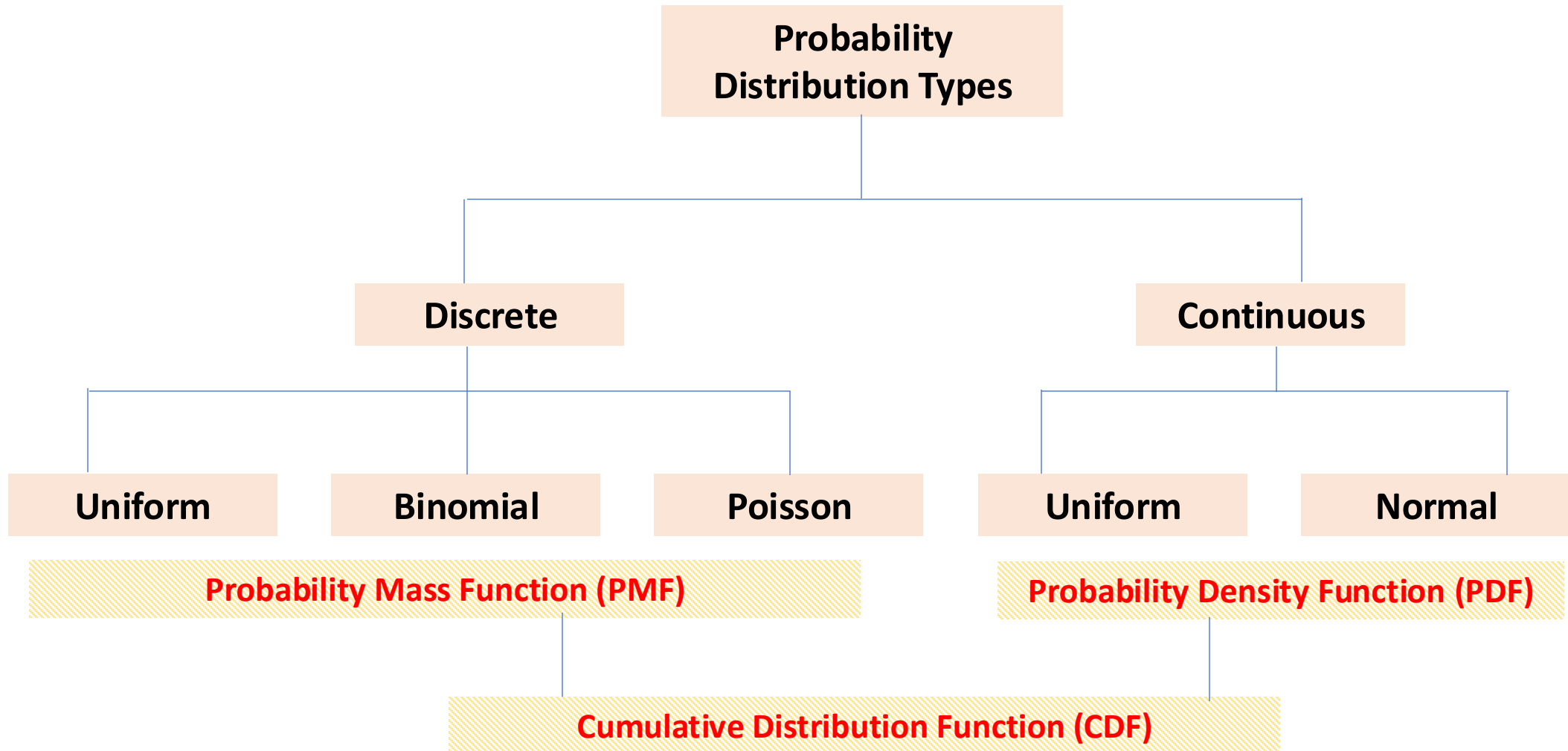
- **Discrete random variable:** Can only take on countable number of possible values
 - Credit ratings: AAA, AA, B, BBB, etc
 - Number of orders received on a shopping website (0, 1, 2, 3, ...)
 - Customer churn: 0 = no churn, 1 = churn
- **Continuous random variable:** Can take on any value in a certain interval
 - Market share of a company (Infinite possible values between 0% and 100%, e.g. 45.123%)
 - Time taken to place an online order (e.g. 2.5 minutes or 2.55 minutes, etc)
 - Waiting time at an ATM (e.g. 30.2 seconds or 45.76 seconds, etc)

Probability Distribution

- **Distribution:** Describes all the probable outcomes of a **variable**
- **Discrete distribution:** Sum of all individual probabilities must equal 1
- **Continuous distribution:** Total area under the **probability curve (density)** equals 1



Probability Distribution Types and Estimations



Discrete Distributions:

(1) Uniform Distribution

- Note: Uniform Distribution can also fall under continuous probability distributions category, depending on what we are measuring. If the observations are not discrete, a uniform distribution becomes continuous.
- Uniform Discrete Distribution: Throw of a dice (exactly 1, 2, ..., 6)
- Uniform Continuous Distribution: Time taken by a flight (can go up to nano seconds and beyond)

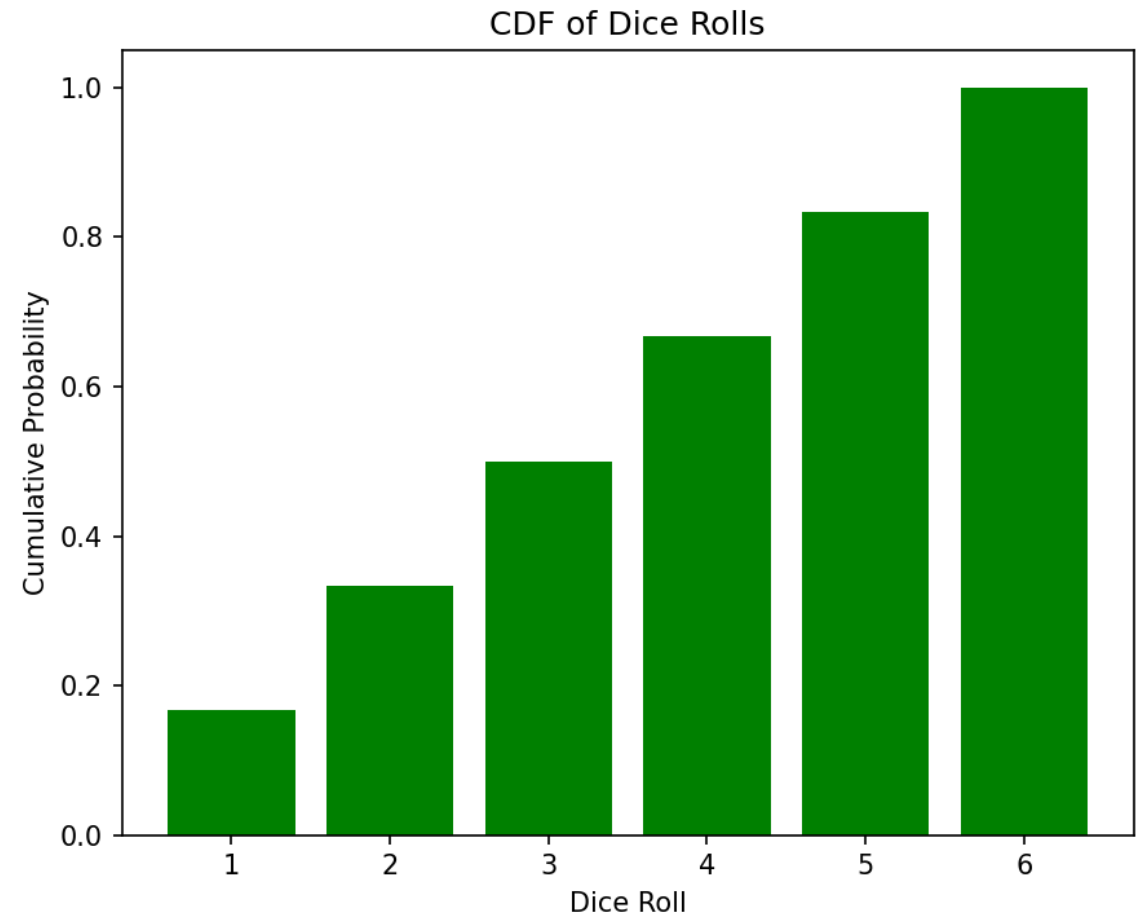
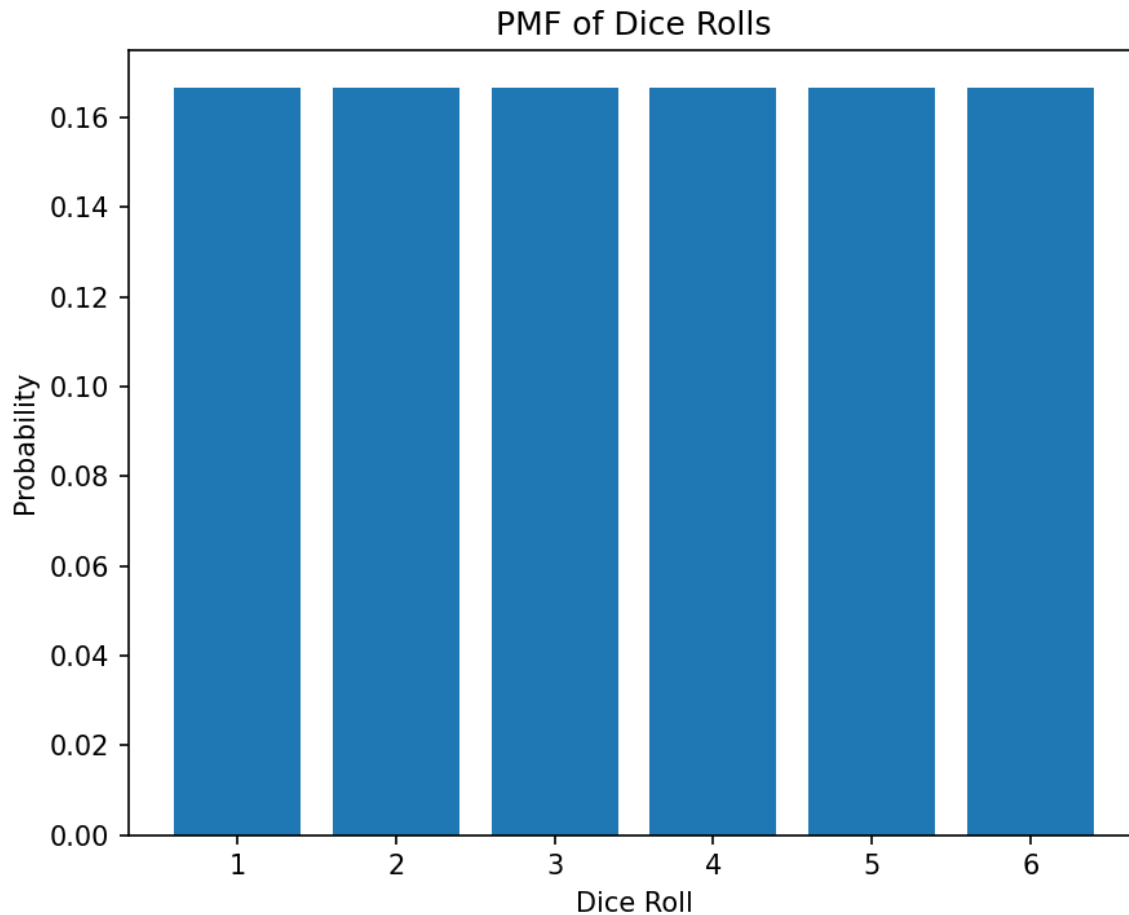
Uniform Distribution



- Rolling a fair die: 6 discrete, equally probable outcomes
- We can roll a 1 or 2, but not 1.5
- The probabilities of each outcome are evenly distributed across the sample space
- Code: `C:\code\Data Analytics\uniform_dist.py`

Uniform Distribution

- Rolling a die has 6 equal probabilities and their total adds to 1



Discrete Distributions:

(2) Binomial Distribution

Binomial Distribution

- **Bernoulli Trial:** An experiment with only two possible outcomes – success or failure
- Example: Heads or Tails, On or Off, Sick or Healthy
- Many such Bernoulli trials = **Binomial distribution**
- So, tossing a coin once = Bernoulli trial, tossing it say 10 times = Binomial distribution
- Binomial distribution: The two outcomes do not always have to be equally probable
 - Example: In a fruit basket, there are 7 apples and 3 oranges
 - $P(A) = \frac{7}{10}$, $P(O) = \frac{3}{10}$

Binomial Distribution Formula

Problem: Calculate the probability that *3 out of 5 students like Python*, when in general 66% students like Python

Number of students liking Python, e.g. 3

Probability that someone likes Python, e.g. 66%

$$\bullet p(x \mid n, p) = \left(\frac{n!}{x!(n-x)!} \right) p^x (1 - p)^{n-x}$$

Probability

Number of students we asked, e.g. 5

Binomial Distribution Example (C:\code\Data Analytics\binom_student.py)

- Problem: Calculate the probability that 3 out of 5 students like Python, when in general 66% students like Python
- $p(x \mid n, p) = \left(\frac{n!}{x!(n-x)!} \right) p^x (1 - p)^{n-x}$
- $p(x = 3 \mid n = 5, 0.66) = \left(\frac{5!}{3!(5-3)!} \right) 0.66^3 (1 - 0.66)^{5-3}$
- $p(x = 3 \mid n = 5, 0.66) = \left(\frac{5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1 (2 \times 1)} \right) (0.66 \times 0.66 \times 0.66) (0.33)^2$
- $p(x = 3 \mid n = 5, 0.66) = \left(\frac{5 \times 4}{2} \right) 0.287 \times 0.115$
- $p(x = 3 \mid n = 5, 0.66) = 10 \times 0.287 \times 0.115$
- $p(x = 3 \mid n = 5, 0.66) = 0.33$ Result: **Binomial Probability Mass Function (PMF)**

Exercise (Solve + Code)

- Calculate the probability that 2 out of 7 students prefer online learning over in-person classes when, in general, 55% of students prefer online learning.
- $p(x \mid n, p) = \left(\frac{n!}{x!(n-x)!} \right) p^x (1 - p)^{n-x}$
- $p(x = 2 \mid n = 7, 0.55) = \left(\frac{7!}{2!(7-2)!} \right) 0.55^2 (1 - 0.55)^{7-2}$
- $p(x = 2 \mid n = 7, 0.55) = \left(\frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 1 (5 \times 4 \times 3 \times 2 \times 1)} \right) (0.55 \times 0.55) (0.45)^5$
- $p(x = 2 \mid n = 7, 0.55) = \left(\frac{42}{2} \right) 0.3025 \times 0.0184$
- $p(x = 2 \mid n = 7, 0.55) = 21 \times 0.3025 \times 0.0184$
- $p(x = 2 \mid n = 7, 0.55) = 0.1168$

Understanding PMF and CDF

- **Probability Mass Function (PMF)**: Probability at a specific point
- **Cumulative Distributed Function (CDF)**: Cumulative probability up to a specific point
- Example: Consider tips dataset
- Suppose we define *success as tip $\geq 15\%$ of the total bill amount*
- **PMF** = Probability for each possible number of successful tips, e.g.
 - $P(X=0)$ -> Probability that none of the tips are at least 15%
 - $P(X=1)$ -> Probability that exactly one tip is at least 15%
 - $P(X=2)$ -> Probability that exactly two tips are at least 15%
 - ... up to the total number of tips
- **CDF** = Cumulative probability at that point
- C:\code\Data Analytics\binomial_distribution_tips_understanding.py

Discrete Distributions:

(3) Poisson Distribution

Poisson Distribution

- Binomial distribution: Perform n trials and determine success/failure
- **Poisson distribution:** Considers the number of successes *per unit of time* (or another continuous unit, such as distance)
- Example: On an average, we read 10 pages of a book in an hour – what are the chances of reading 8 pages in the next hour?
- $$P(x) = \frac{\lambda^x e^{-\lambda}}{x!} = \frac{10^8 e^{-10}}{8!} = \frac{10^8 \times 2.718^{-10}}{8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1} = 0.113 \text{ or } 11\%$$
- Note: $e = 2.718$, Euler's constant

Poisson Distribution

- Example: Usually 15 cars pass a toll plaza every hour. We want to know the probability that 13 cars will pass through in the next hour.

- $$P(x) = \frac{\lambda^x e^{-\lambda}}{x!} = \frac{15^{13} 2.718^{-15}}{13!} = \frac{10^8 x 2.718^{-10}}{8x7x6x5x4x3x2x1} = 0.0956 \text{ or } 9.56\%$$

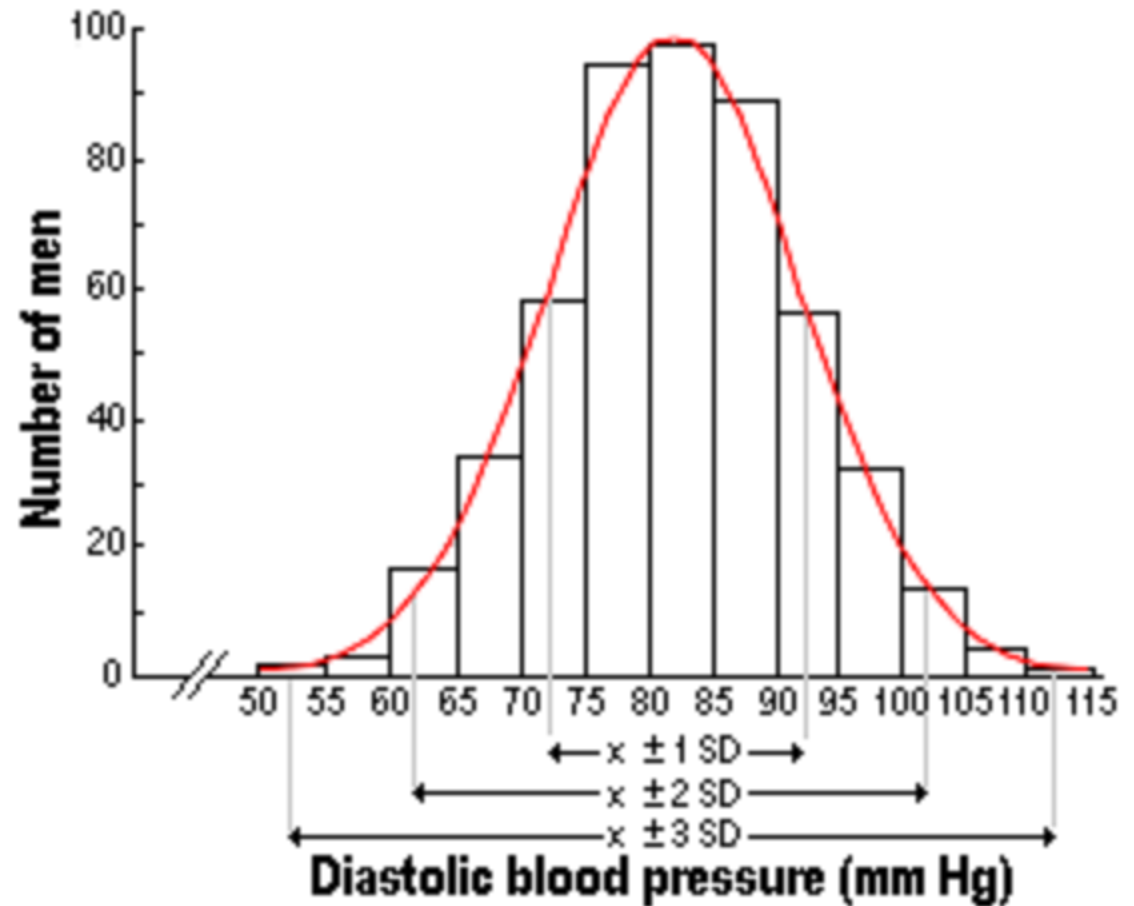
Poisson Distribution – Code

- C:\code\Data Analytics\poisson_distribution.py
- C:\code\Data Analytics\poisson-retail-transactions.py
- Assignment: For Amazon Sale Report data:
 - Find the average monthly items sold for January to December separately
 - For each month, find the probability of selling the average ± 5 products using Poisson distribution
 - Visualize results

Continuous Distributions: Normal Distribution

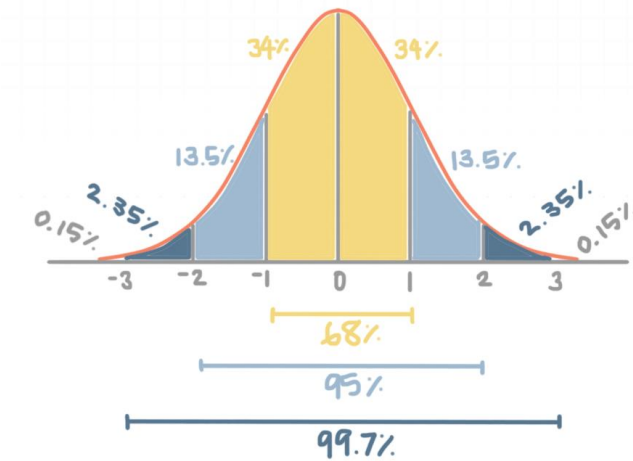
Real life and Normal Distribution

- Many real-life data points follow **Normal Distribution**:
 - People's heights and weights
 - Population blood pressure
 - Test scores
- Also called as **Gaussian Distribution**
- Generally, less/non-natural phenomena do not have normal distributions, e.g. income of people

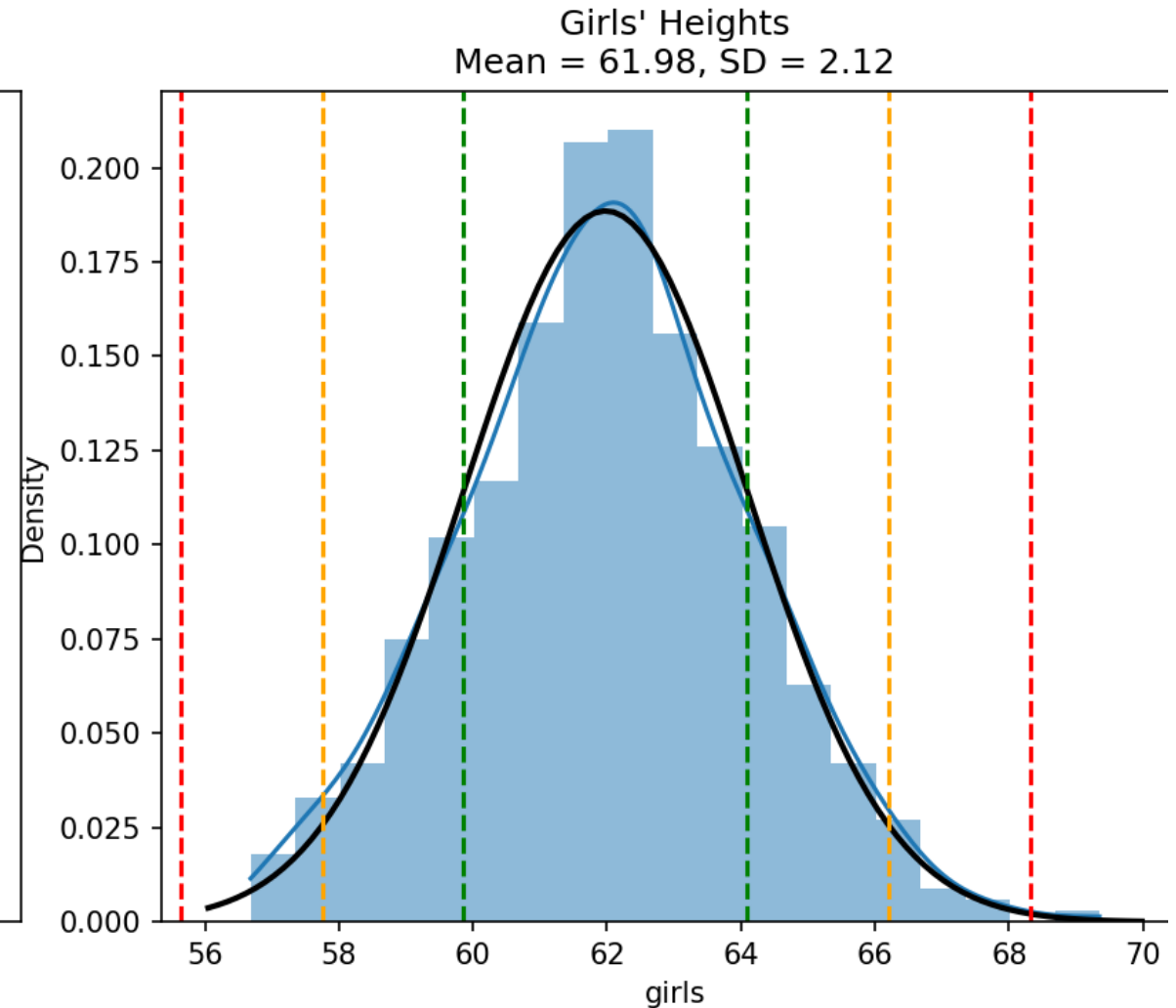
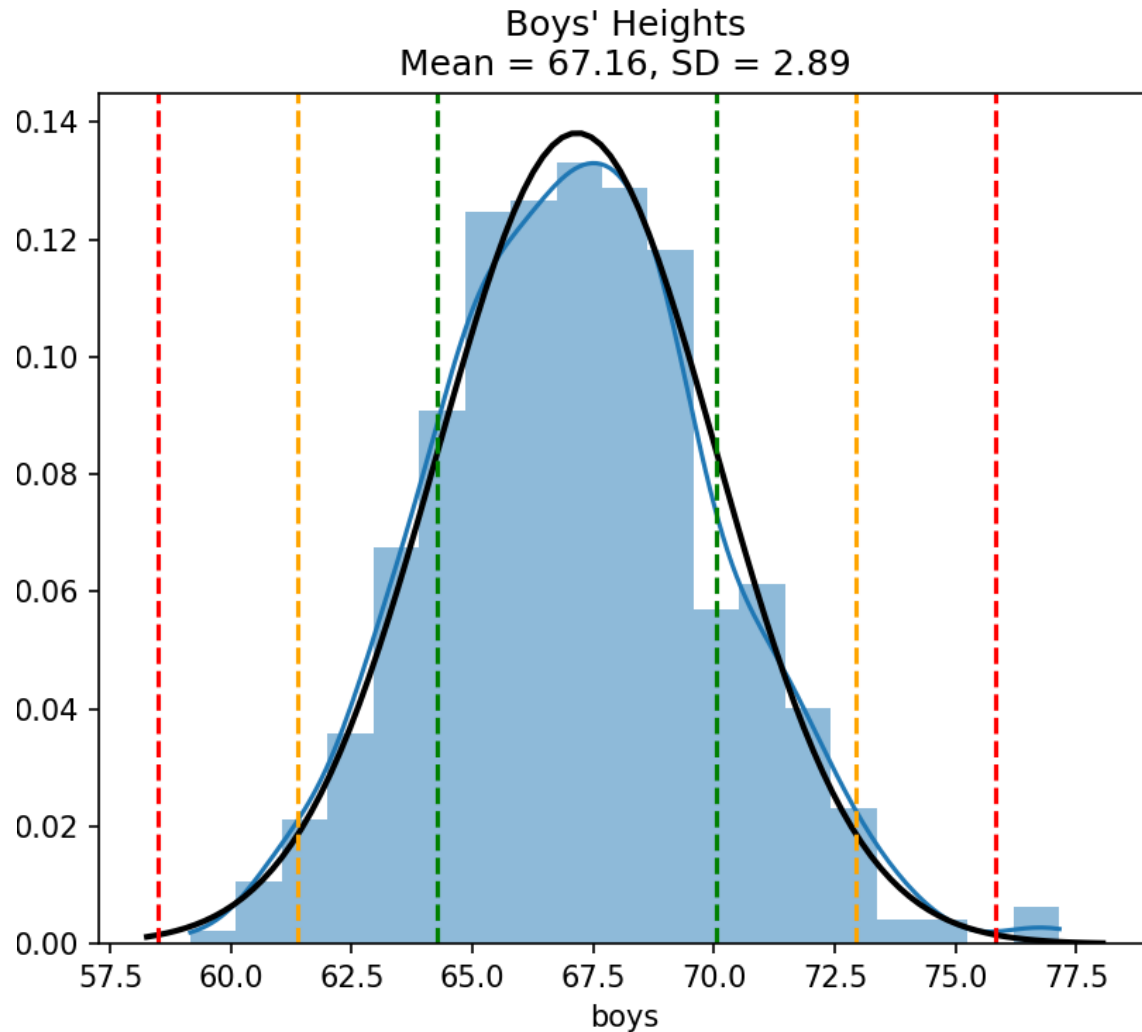


Normal Distribution: Key Features

- **Symmetry:** Perfect symmetry around the mean
- Mean = Median = Mode = Center point of the normal distribution
- **Bell-shaped curve**
- **Empirical rule:**
 - 68% of the data falls within $\text{Mean} \pm 1 \text{ SD}$
 - 95% of the data falls within $\text{Mean} \pm 2 \text{ SD}$
 - 99.7% of the data falls within $\text{Mean} \pm 3 \text{ SD}$
- **Z-score (Standard score):** Useful in finding **relative position** of an observation with respect to the overall population



Example: Student Heights (C:\code\Data Analytics\normal_distribution.py)

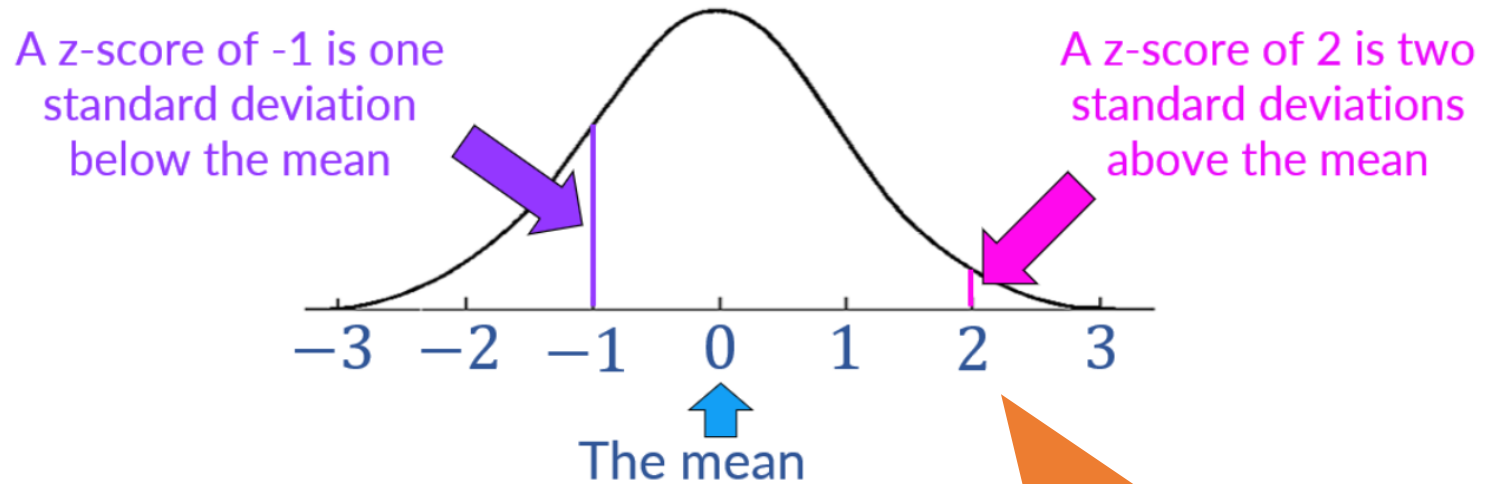


Z-Score Calculations

- **Z-Score (Standard Score):** Describes *how many standard deviations away* a data point is *from the mean*

- **Z-Score** =
$$\frac{x - \mu}{\sigma}$$

- x : individual data value
- μ : population mean
- σ : population standard deviation



Z-scores $> +3$ and < -3 are considered outliers

Z-Score: Problem

- A runner participated in a 200m race and a 500m race
- Consider the following, calculate Z-scores and determine where she did better

Race	Average time	Standard deviation	Runner's time
200m	31s	1.5s	28s
500m	125s	8.2s	132s

Z-Score Example

- A runner participated in a 200m race and a 500m race
- Consider the following and determine where she did better

Race	Average time	Standard deviation	Runner's time
200m	31s	1.5s	28s
500m	125s	8.2s	132s

200m

$$Z = \frac{X - \mu}{\sigma} = \frac{28 - 31}{1.5} = -2$$

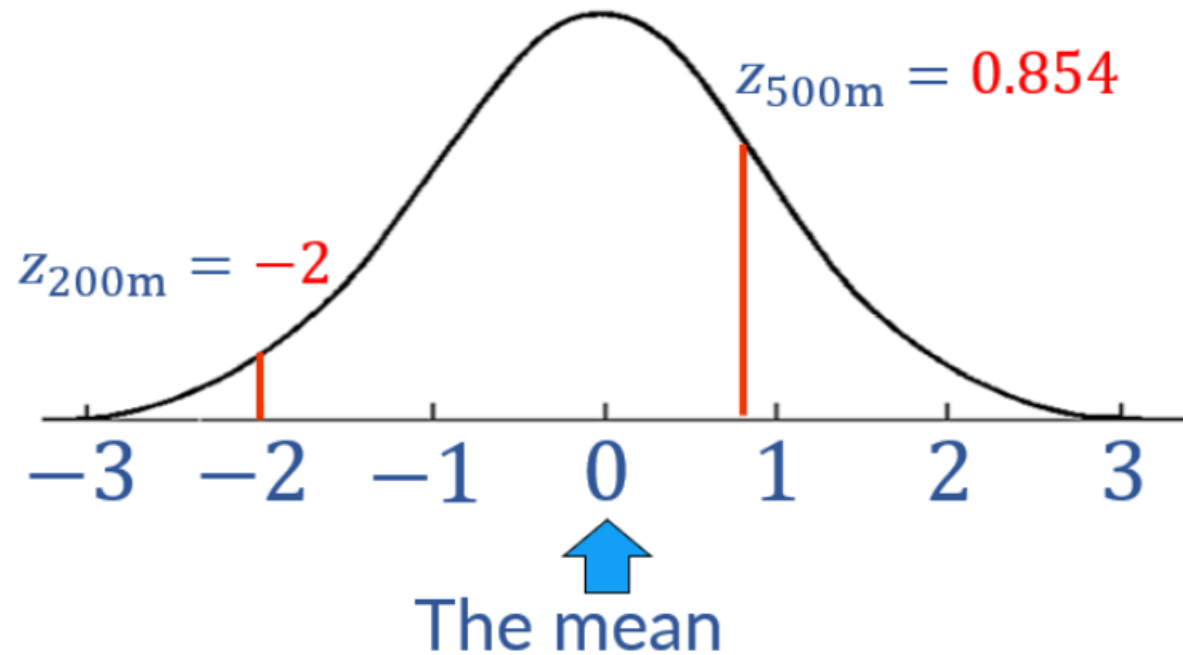
500m

$$Z = \frac{X - \mu}{\sigma} = \frac{132 - 125}{8.2} = 0.854$$

Visualizing Z-Scores

In other examples, positive/higher Z-score will be better, e.g. marks obtained by a student –
Because, here the student would want to be *above* average

- In this example, a lower time would be preferable when completing a race and so, the lower z-score would be better



The 200m time was below the mean and was therefore the quicker than the average runner

Student Example: Z-Scores, Probabilities, Percentiles

- Boy with a height of 62 inches
- Z-Score = $\frac{62 - 67.16}{2.88} = -1.79$
- From **negative scores** Z-table (Row = -1.7, Column = 0.09), Probability = 0.0367
- This student is in the 3.67th percentile
 - He is taller than 3.67% boys
 - 96.33% boys are taller than him
- Girl with a height of 63 inches
- Z-Score = $\frac{63 - 61.98}{2.11} = 0.48$
- From **positive scores** Z-table (Row = 0.4, Column = 0.08), Probability = 0.6844
- This student is in the 68.44th percentile
 - She is taller than 68.44% girls
 - Only 31.56% girls are taller than her

Three Important Measurements

- Dataset: diabetes.csv, mean Glucose (μ) = 120.89, SD (σ) = 31.97
- **Relative position (Z-score)** = $\frac{x - \mu}{\sigma}$
- **Absolute value (x)** = (z-score * σ) + μ
- **Percentile at the Z-score** = CDF at that Z-score
- If we know Z-score, we can find percentile, and vice versa, using Z-table

Examples (z_score_diabetes.py)

- Dataset: diabetes.csv, mean Glucose = 120.89, SD = 31.97

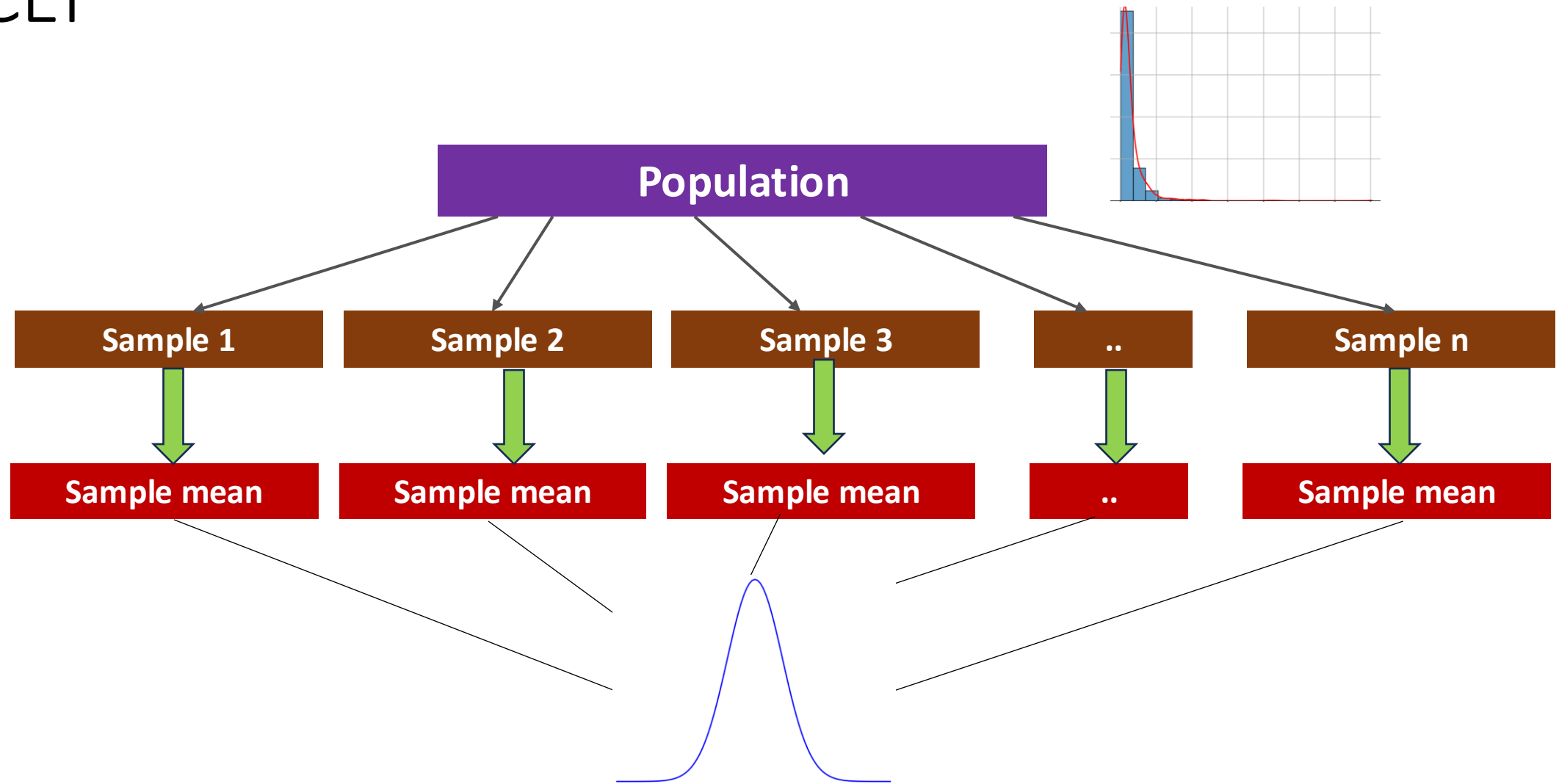
What we know	What we want to find out	Formula	Actual values
x ... A patient has Glucose = 168	Z-score	$z\text{-score} = \frac{x - \mu}{\sigma} = \frac{168 - 120.89}{31.97}$	z-score = 1.47
	Percentile	Search the Z-score (1.47) in the Z-table to get the Cumulative probability up to that Z-score	Percentile = 0.9292
Percentile ... A patient is at the 30 th percentile	Z-score	Go to Z-table and search for cumulative probability = 0.3000 to get the Z-score	Z-score = -0.52
	Corresponding Glucose value	$x = (z\text{-score} * \sigma) + \mu$	Glucose = 104.12
Z-score ... A patient's z-score = 2.8	Corresponding Glucose value	$x = (z\text{-score} * \sigma) + \mu$	Glucose = 210.41
	Percentile	Search the Z-score (2.8) in the Z-table to get the Cumulative probability up to that Z-score	Percentile = 0.9974

The Central Limit Theorem (CLT)

Central Limit Theorem (CLT)

- Problem: Suppose population data does not follow normal distribution (i.e. it is left/right-skewed)
- Population->Samples
- Example: 10 lakh examination result of students->500 samples of 100 students each
- For each sample, calculate average marks (Sample mean or $\bar{x}$)
- Plot these sample means on a graph
- They will follow normal distribution: **Central Limit Theorem (CLT)**
- Generally, minimum sample size = 30
- How many such samples? No such number
- Result: Consider original population also as normally distributed now

CLT



Sampling from a Normal Population

- Suppose we know that a population is normally distributed with mean = μ and SD = σ
- From this, we can take an unlimited number of samples of size n
- For each sample, we calculate mean and plot all these sample means
- Now we compute mean of sample means = $\mu_{\bar{x}}$ and SD of sample means = $\sigma_{\bar{x}}$
- We will find that $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}}$ (called **Standard error**) = $\frac{\sigma}{\sqrt{n}}$
- Conclusions
 - Mean of *means of samples drawn from a population* = Population mean
 - As n becomes larger, standard error will reduce

Z-Score Calculation

- When sampling from a population, the Z-score formula changes
- When calculating for population data: $z\text{-score} = \frac{x - \mu}{\sigma}$
- When calculating for sampling from a population: $Z\text{-score} = \frac{x - \mu}{\frac{\sigma}{\sqrt{n}}}$
- Reason: We are working with a sample, rather than an entire population, so we wish to take a smaller standard error (standard deviation)

More the Samples, Lesser the Error

- Suppose a manufacturer ships goods in a box so that the weights of the boxes are normally distributed with a mean weight of 25 kg with a standard deviation of 4 kg. What is the impact of various sample sizes on the Z-score calculation, and hence the estimation?
- Solution: Next slide
- **Key points**
 - **Precision of estimates:** If we take a sample of (a) 5 boxes and calculate the average, (b) 100 boxes and calculate the average ... option (b) will always give us more accurate results
 - **Law of large numbers:** With small samples, we have a risk of randomness/bad samples
 - **CLT:** As we have more samples, each having a larger sample size, the distribution of sample means approaches normal distribution

More Samples, Better Results

- As the sample size increases, the probability that a box will weigh between 24 kg and 27 kg becomes even higher due to the reduced variability in the sample mean. This makes our predictions more accurate.

Sample size	Lower range Z-score	Higher range Z-score	Using Z-table, CDF	Probability range
n	$Z\text{-score} = \frac{x - \mu}{\frac{\sigma}{\sqrt{n}}}$	$Z\text{-score} = \frac{x - \mu}{\frac{\sigma}{\sqrt{n}}}$		
1	$\frac{24 - 25}{\frac{4}{\sqrt{1}}} = \frac{-1}{4} = -0.25$	$\frac{27 - 25}{\frac{4}{\sqrt{1}}} = \frac{2}{4} = 0.50$	0.4013 (for -0.25) to 0.6915 (for 0.50)	0.6915 – 0.4013 = 0.2902 or 29%
25	$\frac{24 - 25}{\frac{4}{\sqrt{25}}} = \frac{-1}{0.8} = -1.25$	$\frac{27 - 25}{\frac{4}{\sqrt{25}}} = \frac{2}{0.8} = 2.50$	0.1056 (for -1.25) to 0.9938 (for 2.50)	0.9938 – 0.1056 = 0.8882 or 89%
50	$\frac{24 - 25}{\frac{4}{\sqrt{50}}} = \frac{-1}{0.566} = -1.77$	$\frac{27 - 25}{\frac{4}{\sqrt{50}}} = \frac{2}{0.566} = 3.53$	0.0384 (for -1.77) to 0.9998 (for 3.53)	0.9998 – 0.0384 = 0.9614 or 96%
100	$\frac{24 - 25}{\frac{4}{\sqrt{100}}} = \frac{-1}{0.4} = -2.5$	$\frac{27 - 25}{\frac{4}{\sqrt{100}}} = \frac{2}{0.4} = 5.00$	0.0062 (for -2.5) to 1 (for Z-score of 5)	1 – 0.0062 = 0.9938 or 99%

Hypothesis Testing

Hypothesis

- **Hypothesis:** Not possible to test entire population, so test sample and draw conclusions about the population (**Inferential statistics**)
- **Hypothesis testing:** Lets us evaluate how well a sample supports an assumption about the population using these steps:
 - State our assumption (**Null hypothesis** and **Alternate hypothesis**)
 - Determine probability of error (**Level of significance**) called α
 - Check how well the data supports our assumption (Calculate **Test statistic**)
 - Translate that into a probability that supports it (**p-value**)
 - Conclusion: Is $p\text{-value} < \alpha$?
 - Yes – Our assumption was wrong – Reject Null hypothesis
 - No – Our assumption was correct – Fail to reject Null hypothesis
 - Alternatively, compare test statistic with **Critical value** and decide
- Sample statistic: $\bar{x}$ (Sample mean) $\rightarrow$ Population parameter: μ (Population mean)

Null and Alternate Hypotheses

- **Null hypothesis:** Neutral statement (*status quo*)
- **Alternate hypothesis:** What we actually want to be true
- Example: Suppose, the chances of a student getting placed before doing a C-DAC course are 40%
- **Null Hypothesis (H_0):** There is no change in the chances of placement for a student after completing a C-DAC course ($\bar{x} = \mu$)
- **Alternate Hypothesis (H_a or H_1):** There is a significant difference ($\bar{x} \neq \mu$)
- Note: We never *prove/accept* a hypothesis; we either *reject* or *do not reject* a hypothesis

Significance Level (α) and p-value

α : What we set before a statistical test
p-value: What we calculate after the test

- **α = Significance level** = Threshold or acceptable probability of rejecting the null hypothesis when it is actually true
- Usually set to 5% (or 0.05)
- Meaning: We are okay with a 5% chance of rejecting the null hypothesis incorrectly
- **p-value** = Result of an actual test (Area of rejection)
- **Low p-value: Reject H_0 , High p-value: Fail to reject H_0**
- So, we may start with $\alpha = 0.05$, run a test, and calculate p-value
 - If p-value = 0.02, Reject H_0
 - If p-value = 0.10, Do not reject H_0

If p-value $\leq \alpha$... Reject H_0
If p-value $> \alpha$... Do not reject H_0

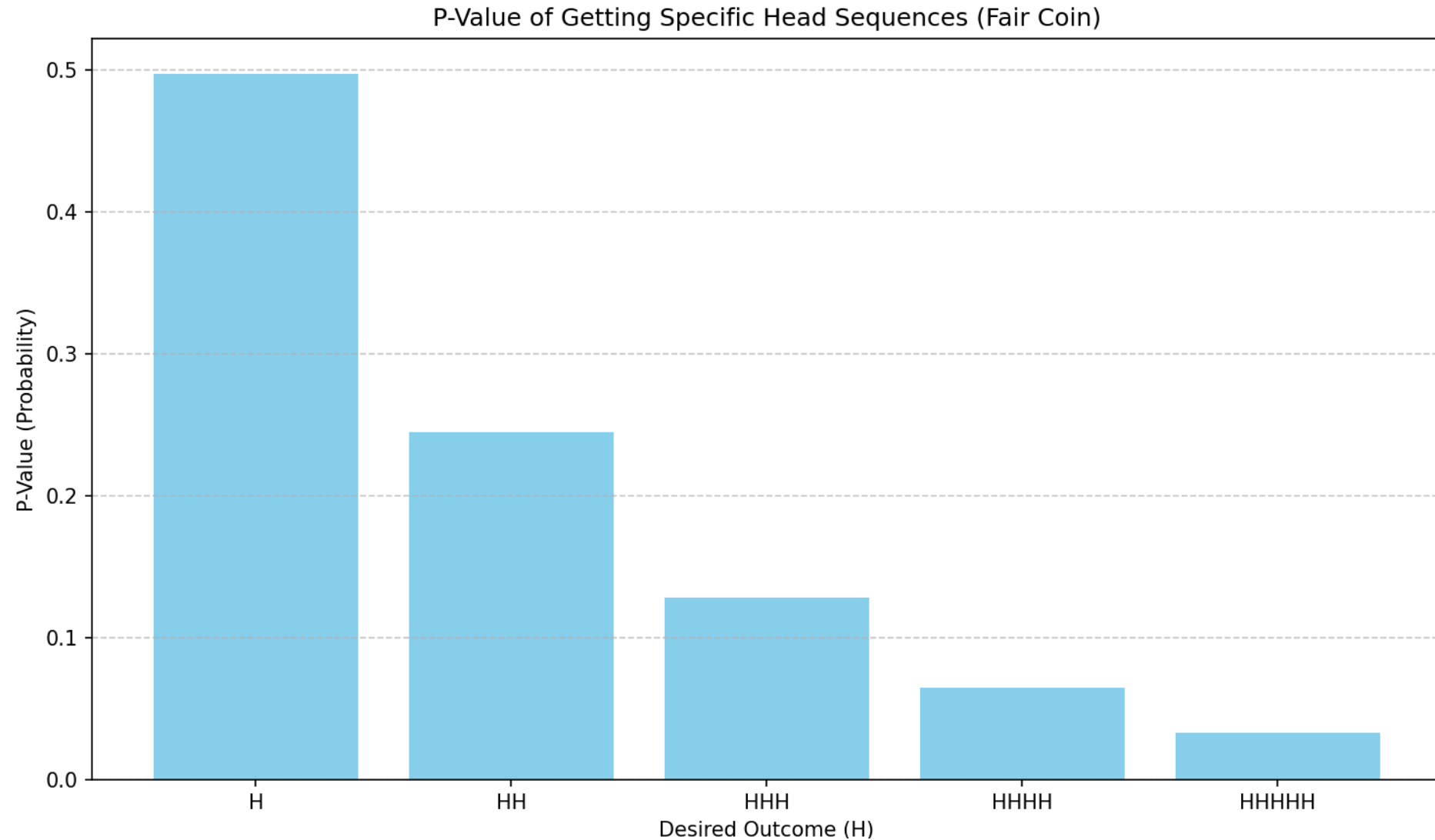
Significance Level (α) and p-value

- Example: Tossing a coin with H_0 = Coin is fair and H_1 : Coin is not fair
- We toss it 5 times and each time we get a head! (See next slide for a bar plot)

Toss Number	Result	Current Situation	Probability formula	Theoretical Probability (p-value)
1	H	H	$\frac{1}{2}$	0.50 (50%)
2	H	HH	$(\frac{1}{2})^2$	0.25 (25%)
3	H	HHH	$(\frac{1}{2})^3$	0.125 (12.50%)
4	H	HHHH	$(\frac{1}{2})^4$	0.06 (6%)
5	H	HHHHH	$(\frac{1}{2})^5$	0.03 (3%)

- At what point we conclude that the coin is not fair and reject H_0 ?
- That is our **significance level (α)**, usually set at 0.05 (i.e. 5%) or 0.10 (Or 10%) – Our *Lakshman Rekha!* Our **p-value** must be $> \alpha$ for us not to reject H_0
- If we have set it at 5%, if we get a heads a fifth time, the p-value $< \alpha$ and we reject H_0

Visualizing p-value

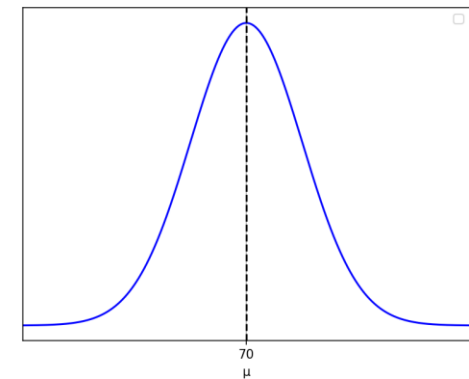


Understanding p-value

- **p-value:** Probability of obtaining a result at least as extreme as the observed one, assuming the null hypothesis is true
- Helps us determine whether an observed result is due to random chance or if it is statistically significant
- Example: Tossing a coin five times and counting the number of heads, repeat this say 100 times
- We would expect HHHHH to occur about 3.125% of the time (i.e. approximately 3 times in 100 such experiments) due to random chance
- But suppose we find that this is happening 10-15 times
- Conclusion: Question if the coin is fair, since we expected it to be only about 3 times , due to random chance

Significance Level (α) – More Details

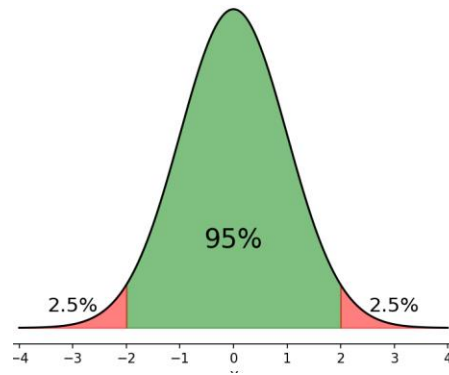
- Interesting point! Another definition of α : The probability of rejecting the null hypothesis even though it is true (**Type I Error**)
- Why? What if the coin was indeed fair and showed HHHHHH? – Very rare but possible!
- We ended up rejecting H_0 when we should not have! => **Type I error**
- Another Example: A course head thinks average student score is 70%
- H_0 : Population mean score (μ) = 70%, H_1 : Population mean score (μ) $\neq$ 70%
- So, scores distribution should look like this



Significance Level

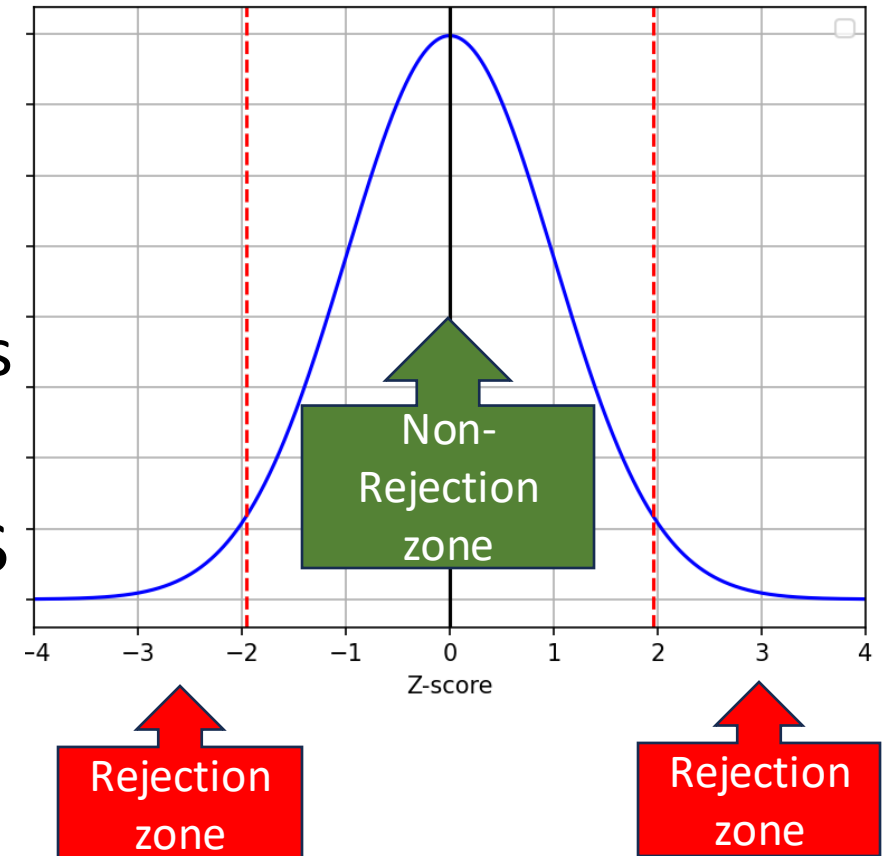
- Now we can take a sample and calculate its sample mean and use it to calculate the Z-score: $Z = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$
- Note: Z-score formula is actually $Z = \frac{\bar{X} - \mu}{\sigma}$, but if we do not know the population standard deviation (σ), then it becomes $Z = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$
- $Z = 0 \Rightarrow$ Sample mean = Hypothesized population mean
- Anything close to a Z-score of 0: Do not reject H_0
- How large should Z-score be, for us to reject H_0 ?

Significance Level

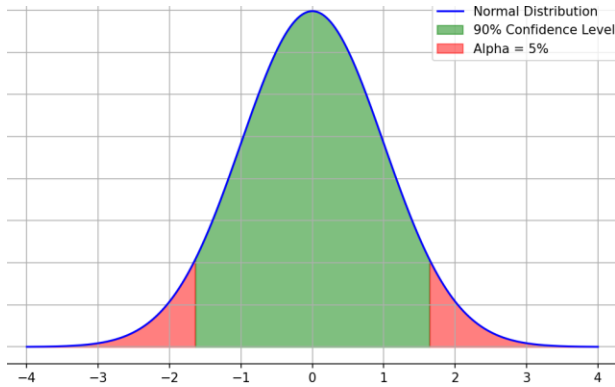


Critical values define the boundaries of the **non-rejection zone** for Z-scores ... If our Z-score crosses this zone on either side, we reject H_0 ...

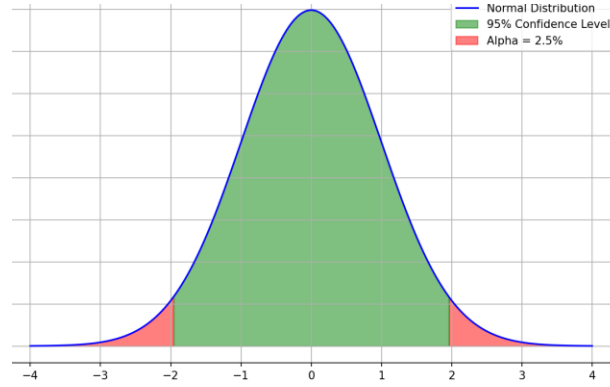
- If Z-score is in the **middle part**, we do not reject H_0
- But if it goes **beyond the dashed red lines**, we reject H_0 – Why?
- Those are **rejection regions (Critical zone)**
- Area of rejection regions depends on **significance level (alpha)**
- If $\alpha = 0.05$, we have 0.0250 on both sides
- From Z-table, when $\alpha = 0.0250$, **critical value** = 1.96 on RHS and -1.96 on LHS
- So, if Z-score < -1.96 or > 1.96 , reject H_0



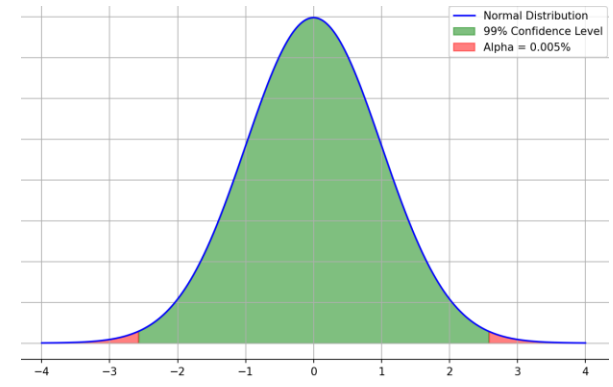
Significance Level and Confidence Level



Confidence level = 90%
Significance level (α) = 10%
Critical Z-Value = ± 1.64



Confidence level = 95%
Significance level (α) = 5%
Critical Z-Value = ± 1.96



Confidence level = 99%
Significance level (α) = 1%
Critical Z-Value = ± 2.58

- As the confidence level increases, Critical Z-value increases

How to Get Critical Value from Z-Table

- Look for our probability value
- That is 0.025
- Critical value = Row + Column
- Critical value = 1.9 + .06
- Critical value = 1.96

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455

- Note: If we look in the *positive* Z-table (the second table on the same page, we will need to look for $1 - 0.025$, i.e. 0.975 and we will get the same result, i.e. 1.96)

Type I, II Errors, Alpha Beta

	H0 is True	H0 is False
Reject H_0	Type I Error , Alpha	Correct
Do not Reject H_0	Correct	Type II Error , Beta

- Why can this happen? Bad sample data ...
- Example: We got a bad sample where Average sample score = 80
- We would reject H_0 because in our sample it seemed so, although in general it is true, considering all the other samples
- Note: Try to reduce Type I errors => Try not to reject H_0 (See table above)
- But it automatically means a scope for increase in Type II errors (See table above)

Visualizing Type I Errors

$$\alpha = .05$$

95% of all sample means ($\bar{x}$) are *hypothesized to be* in this region.

Fail to reject null hypothesis

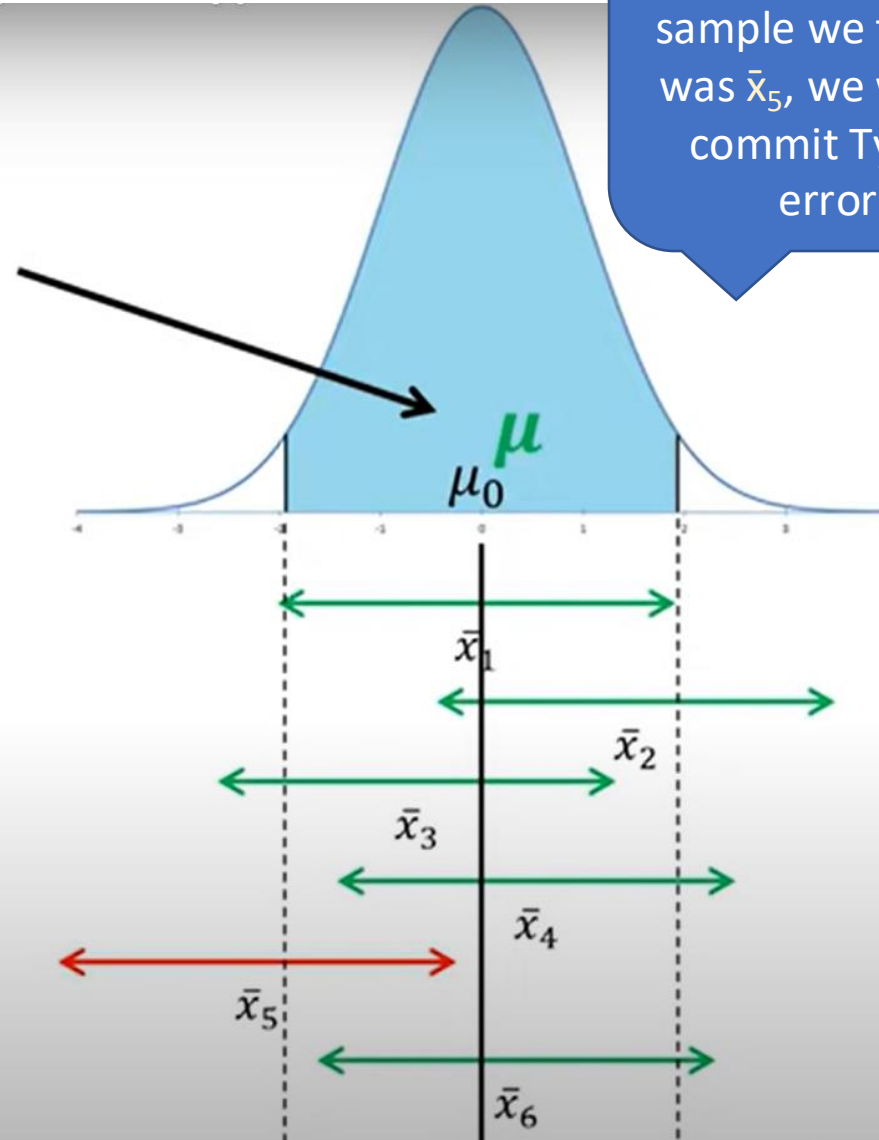
Fail to reject null hypothesis

Fail to reject null hypothesis

Fail to reject null hypothesis

Reject null hypothesis

Fail to reject null hypothesis



Meaning: If by chance, the only sample we tested was $\bar{x}_5$, we would commit Type I error

According to all the other samples, we will fail to reject H_0 , which is correct

$$H_0: \mu = \mu_0$$

$$H_a: \mu \neq \mu_0$$

Type I error

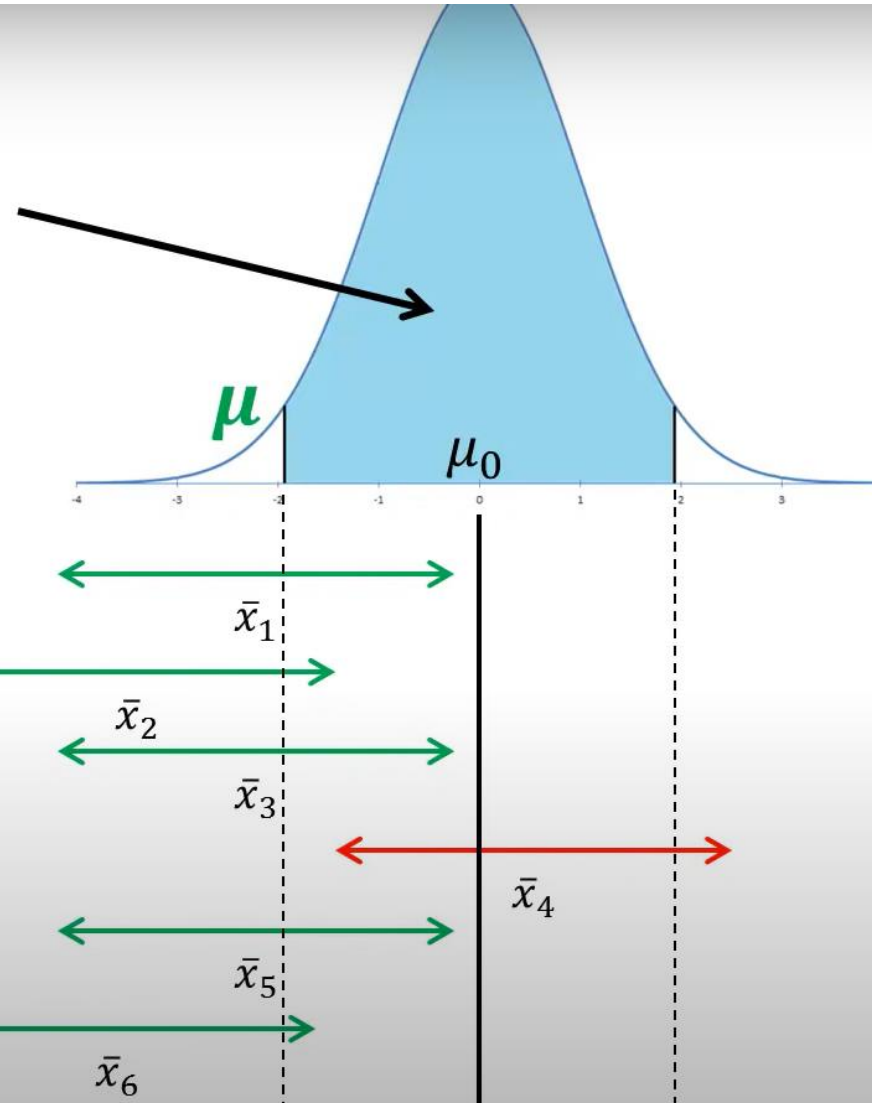
If we took a sample and it was by chance like $\bar{x}_5$, we would **incorrectly reject the null hypothesis.**



Similarly, Type II Error Now

According to all the other samples, we will reject H_0 , which is correct

95% of all sample means ($\bar{x}$) are *hypothesized to be* in this region.
 $\alpha = .05$



Reject null hypothesis

Reject null hypothesis

Reject null hypothesis

Fail to reject null hypothesis

Reject null hypothesis

Reject null hypothesis

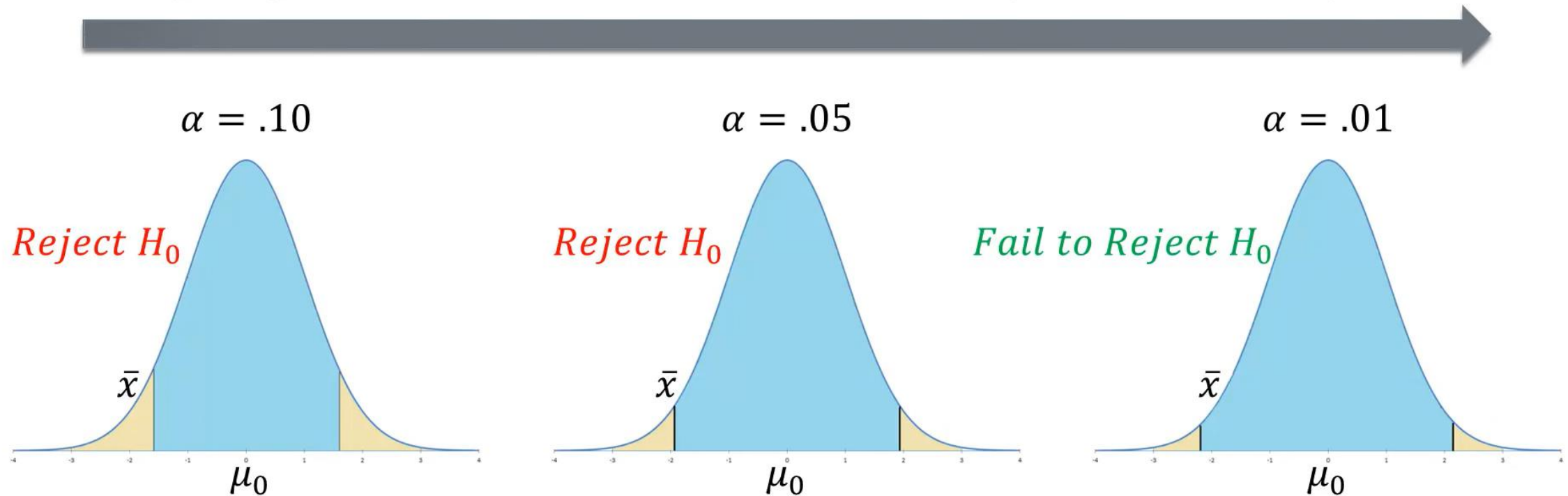
$$H_0: \mu = \mu_0$$

$$H_a: \mu \neq \mu_0$$

If we took a sample and it was by chance like $\bar{x}_4$, we would **incorrectly “accept” the null hypothesis.**

Meaning: If by chance, the only sample we tested was $\bar{x}_4$, we would commit Type II error

As α decreases so does the chance of Type I error. The critical value to reject the null hypothesis moves outward thus “capturing” more sample means. Kind of like moving the goal posts on a football pitch; the wider they are, more kicks go in.



However the move outward of the critical values may also “capture” a mean from a different population off to the side. We would fail to reject the null when indeed we should. Thus the chance of Type II error increases as α decreases.

Calculate Test Statistic

- Suppose to test our earlier H0: Sample size = 50, Sample average ($\bar{x}$) = 67 inches, SD (σ) = 5 inches
- $Z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{67 - 65}{\frac{5}{\sqrt{50}}} = 2.82$
- Alpha = 0.05, but since this is a two-tailed test, we need to consider alpha / 2 = 0.05 = 0.025
- See in Z-table (<https://www.z-table.com/>) the probability value of 0.025 to get the critical value (i.e. expected z-score as per normal distribution) ... See next slide
- Critical value = 1.96 (i.e. Up to ± 1.96 of Z-score, we would not reject H_0)
- Since our calculated test statistic > critical value, we reject H0

p-value

- **p-value** = Area of rejection
- Low p-value: Reject H_0 , High p-value: Fail to reject H_0
- Example: A pizza outlet claims they deliver pizza in ≤ 30 minutes on average
- H_0 : Mean delivery time = 30 minutes
- H_a : Mean delivery time $\neq 30$ minutes
- We test a few sample delivery times and get a p-value = 0.001
- Since 0.001 is much less than 0.05, we reject H_0

p-value

- **p-value** = Area of rejection
- Low p-value: Reject H0, High p-value: Fail to reject H0
- In our example, Z-score = 2.82
 - Go to Z-table and look for 2.82 ... We see 0.9976 ... So, this Z-score is at the 99th percentile ... Which means only $1 - 0.9976 = 0.0024$ or 0.24% values are above it ... Here p-value = 0.0024 ... But since it is a two-tailed test, we need to take $2 \times \text{p-value} = 2 \times 0.0024 = 0.0048$
 - We can also use a ready calculator to get the same result:
<https://www.socscistatistics.com/pvalues/normaldistribution.aspx>
- Interpretation: If our null hypothesis were true we would expect to see a Z-value of 2.82 only in 0.48% of cases just due to random sampling variability
- Since p-value of 0.0048 < alpha of 0.05, we reject H0

Conclusion

- Fail to reject H_0

Critical Values for Different Levels of α

Approximate Critical Values			
α	Left-Tailed Test	Right-Tailed Test	Two-Tailed Test
0.1	-1.28	+1.28	-1.64 and + 1.64
0.05	-1.64	+1.64	-1.96 and + 1.96
0.01	-2.33	+2.33	-2.58 and + 2.58